



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

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**DIGITAL TRANSFORMATION & ENTERPRISE SYSTEMS AT  
UNIVERSITY TUN HUSSEIN ONN MALAYSIA (UTHM)**

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**Sub System** : Campus transport & Shuttle Management System

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## **ABSTRACT**

Universiti Tun Hussein Onn Malaysia (UTHM) is currently experiencing substantial inefficiencies in its campus transportation operations, which run independently of the university's main digital systems. This fragmentation encourages the use of manual data gathering methods, resulting in data silos, redundancy, and a lack of important functionality for the campus community, such as real-time tracking and occupancy monitoring. As a result, administrators struggle with resource efficiency, and students lack the timely information required for seamless campus movement.

To solve these difficulties, this project presents an integrated Campus Transport & Shuttle Management System built on Enterprise Architecture (EA) principles and the Waterfall Methodology. The system, designed in Figma, interfaces directly to UTHM's master databases and includes six complete modules, including Route Display, Live Shuttle Tracking, ETA Prediction, and an Analytics Dashboard. This architecture bridges the gap between old systems and new operational needs by eliminating data silos and allowing smooth information flow.

This unified digital solution contributes considerably to UTHM's "Smart Campus" concept by replacing manual operations with real-time, data-driven insights. It improves the student experience by enabling predictable trip planning and providing administrators with the analytics needed for successful decision-making and route optimization. Finally, the project delivers a scalable, interoperable architecture that assures UTHM's infrastructure is ready for future digital transformations.

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## CHAPTER 1 : INTRODUCTION

### 1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is a public university located in Batu Pahat, Johor, containing about 18,000 students and staff. Currently, UTHM operates their digital system through several core systems including Student Information System (SMP) for academic related activities, Staff Information Systems (SMS) for employee management, and MyUTHM mobile application for more seamless and efficient user experiences. While it already supports functions that are crucial to run and maintain, the system itself does not fully support areas that include hostel management, student activities, and transport services where these university operations still function in silos or rely on manual interventions. This project focuses on designing an integrated Enterprise System to bridge these gaps, leveraging Figma to support UTHM's vision of a "Smart Campus" mainly on the **Campus transport & Shuttle Management System**.

### 1.2 Problem Statement

Despite having established digital platforms, UTHM faces some operational inefficiencies due to lack of integration between some of the systems with the currently digitalized core systems. The problems include :

- **System fragmentation:** Critical operational services, such as Campus transport & Shuttle Management System, are not fully integrated into digital systems making it harder to connect and update information with legacy systems such as digitalized central Student Information System (SMP), leading to inefficiency in workflow.
- **Data Silos and Redundancy:** Student data is often duplicated across different operational units, causing inconsistencies and data integrity issues.
- **Slow Processing Time:** Handling works manually through Google Forms and Excel sheets requires lots of time which can halt the information processing time.

### 1.3 Project Objectives

The objectives of the project are:

- **Designing an Enterprise Architecture** that integrates all the operational modules and connects it to UTHM's master database.
- **Developing a prototype using Figma** to represent the operational modules for Campus transport & Shuttle Management System and how it will work on a digitalized platform.
- **Creating a data model that eliminates data silos** by connecting the flow of information between the operational modules and the current existing digital system (SMP/SMS).

### 1.4 Project Scope

The scope of the project is to design the Enterprise Architecture that include five key modules including Subject Registration, Academic Advising, Hostel Management, Student Clubs, and Campus Transport. However, for the system implementation, it only focuses on Module 5 : Campus Transport & Shuttle Management System. Under the module includes six sub-module that encompasses Route & Schedule Display, Live Shuttle Tracking, ETA Predictor, Shuttle Occupancy Monitoring, User Feedback, and Transport Analytics.

### 1.5 Project Importance

For UTHM, this project addresses the digital gaps between systems in its operational modules. By transitioning from manual methods to an integrated enterprise system solution, the university can improve their decision making based on real-time information and analytics, such as tracking shuttle vacancies and live location, etc.

For students, the system offers a user-friendly experience that simplifies campus life, from registering for subjects to tracking buses. Ultimately, this architecture serves as a guided blueprint for UTHM's continued digital transformation, ensuring scalability and interoperability for future development.

## **CHAPTER 2 : LITERATURE REVIEW**

### **2.1 Introduction**

Enterprise systems have become essential especially in higher education institutions that oversee intricate administrative and academic processes in order to support digital transformation activities. Universities use a variety of information systems to support teaching, learning, student services, staff administration and infrastructure operations. However, improper system integration frequently leads to disjointed workflows, redundant data and ineffective decision-making.

### **2.2 Enterprise Architecture (EA)**

#### **2.2.1 Definition and Concept of EA**

Enterprise Architecture (EA) is a comprehensive framework that aligns business processes, information flows, applications and technology infrastructure to describe an organization's structure and functioning. The literature claims that EA acts as a guide for enterprise system development, integration and governance in order to guarantee consistency throughout all organizational units.

According to a number of studies, EA is a strategic planning tool that aids businesses in managing complexity by establishing a clear connection between IT resources and business objectives. EA emphasizes an enterprise-wide viewpoint rather than concentrating only on individual systems, making sure that all systems collaborate smoothly to achieve organizational goals.

The core concept of enterprise architecture is typically structured into four main layers:

#### **1. Business Architecture**

This layer outlines the stakeholders, policies, business procedures and function of the organization. Academic operations, student services, administrative

procedure and support services like campus transportation are all included in a university setting.

## **2. Application Architecture**

The systems and applications that support business processes are described at this tier. It focuses on how various applications communicate and exchange data with one another. Systems such as the Staff Management (SMS), the Student Portal (SMP) and the projected Campus Transport and Shuttle Management System are examples of these for UTHM.

## **3. Data Architecture**

Data Architecture specifies the methods for gathering, storing, managing and sharing data amongst systems. By guaranteeing data consistency, integrity and accessibility throughout the organization, EA seeks to remove data silos. In higher education institutions where operational and student data are frequently duplicated across various systems where this is very crucial.

## **4. Technology Architecture**

This layer described the hardware, software platforms, networks and technical infrastructure that support data and applications. It also includes databases and cloud platforms.

An important concept within EA is alignment where IT investments are planned based on corporate strategy rather than isolated system development. Additionally, EA facilitates interoperability which needs smooth communication between new and old systems and scalability which enables systems to change as organizational demands increase.



By facilitating real-time data sharing, integrated services and data-driven decision-making. EA significantly contributes to smart campus projects at higher education institutions. Universities may move from disjointed, manual procedures to a cohesive digital ecosystem that improves user experience, service quality and operational efficiency by implementing EA.

### **2.2.2 Review of previous studies related to EA**

The previous and current research shows that universities around the world are increasingly adopting Enterprise Architecture into their digital system to manage complex workflow and operations as well as supporting “Smart Campus” goals. There are several significant studies in this studies including:

A study by researchers published in MDPI (2023), says that in common, universities often struggle with fragmented flow of information which can come in various factors including the implementation of IT systems without consideration between departments available . Therefore, A study by Adila & Setyawan (2023) suggests that the integration of an EA reference module helps educational institutions align their investments of technology with strategic long term digital development plans. This is relevant for UTHM, where systems like SMP and SIS currently operate in silos, therefore leading to data duplication.

According to recent studies of The Role of Enterprise Architecture (IJERD, 2024), EA acts as a blueprint that outlines how an institution's various IT systems should interact and manage to solve the occurring digital integration and security challenges including fragmentation said earlier. EA can help educational institutions in building a proper protocol for security and data sharing across all systems that ensures the protection of data at each point of their lifecycles. This approach is crucial for UTHM to transition from fragmented methods, such as using Excel or Google Forms, into a unified digital environment.

In a practical case study, researchers applied TOGAF Architecture Development Method (ADM)

To design an integrated IT infrastructure in Nusa Nipa University () to overcome operational efficiencies. The study identifies that the university's current academic governance lacks a structured architecture, which makes the existing IT infrastructure insufficient for responding promptly to business and application changes. Through TOGAF ADM, they first assess the scope of the project to propose architecture that is aligned with the university's vision and mission. Then, focusing on the core architecture layers that include Business, Data, Application, and Technology layers to map out the flow of data across different business processes. The research ended with the implementation of recommendations for migration and later acted as a blueprint. This guide is important for UTHM in order to map subsystems that need to be integrated with existing systems such as SSO gateways and master databases.

## 2.3 Related Sub-System

### 2.3.1 Campus Transport & Shuttle Management System

The Campus Transport & Shuttle Management System is designed to improve how transportation services are managed and accessed within Universiti Tun Hussein Onn Malaysia (UTHM). At present, many transport-related activities rely on manual processes and static information, which can cause delays, confusion, and inefficient communication between students and transport administrators. This sub-system aims to provide a more organized and reliable digital solution that supports daily campus mobility.

For students, the system offers several useful features that help them plan their travel around campus. **The Route and Schedule Display** module allows users to view shuttle routes, stops, and operating times in one place, ensuring that information is clear and easy to access. The **Live Shuttle Tracking** feature enables students to see the current location of shuttles, reducing uncertainty while waiting. In addition, the **Estimated Time of Arrival (ETA) Predictor** helps users estimate when a shuttle will reach a specific stop, allowing better time management.

The system also includes a **Shuttle Occupancy Monitoring module**, which shows the current passenger level of each shuttle. This helps students decide whether to board a shuttle or wait for the next one, improving comfort and reducing overcrowding. To encourage communication and service improvement, the **User Feedback module** allows students to submit complaints or suggestions through a structured platform instead of informal reporting methods.

From the administrative side, the **Transport Analytics Dashboard** provides an overview of shuttle operations, including usage patterns, peak hours, and feedback trends. This information supports better planning and helps administrators identify areas that need improvement. Administrators can also review feedback submissions and update their resolution status to ensure proper follow-up.

Overall, this sub-system provides a more structured and efficient way to manage campus transportation. By combining real-time information, user interaction, and basic analytics, it helps improve daily transport operations and contributes to UTHM's broader goal of building a more connected and digital campus environment.

### **2.3.2 Existing Campus Transport Sub-System at UTHM**

Before the proposed Enterprise-level Campus Transport and Shuttle Management System, UTHM managed its campus shuttle services using a standalone transport tracking platform called the UTHM-KATSANA system. The Student Housing and Transportation Centre administers this system. It offers students access to shuttle routes, schedules, and basic real-time GPS-based bus location information through a web-based interface.

Even though it provides location tracking, the current transport system works separately from UTHM's main information systems like the Student Management Portal (SMP) and the Staff Management System (SMS). This separation means that transport-related data is not combined with student or staff records. As a result, there are fragmented systems and data silos. This isolation prevents centralized data sharing and limits the use of transport information for planning and decision-making.

Additionally, the previous system does not allow for real-time monitoring of shuttle occupancy, predictive ETA analytics, or performance dashboards. Studies on UTHM's shuttle operations show that data on passenger boarding, alighting, and long waiting times were collected using manual methods, like tally counters and stopwatches, instead of being automatically recorded by the system. This reveals a lack of built-in analytical and decision-support features.

For service feedback, user complaints and incident monitoring were usually addressed through separate administrative channels, such as manual records and CCTV reviews. They did not use an integrated digital feedback system. Therefore, while the existing transport system works at a basic level, it lacks the structure, compatibility, and

data-driven insights needed to fully support UTHM's Smart Campus and digital transformation efforts.

## **2.4 Technology Used**

The Campus Transport & Shuttle Management System is designed using Figma, a cloud-based UI/UX design and prototyping tool. Figma is used to create wireframes, user interface layouts, and interactive prototypes that visually represent the system's structure and functionality. Through Figma, the design of key modules such as Route & Schedule Display, Live Shuttle Tracking, ETA visualization, Shuttle Occupancy Monitoring, User Feedback & Complaint Reporting, and the Transport Analytics Dashboard can be clearly illustrated without requiring actual system implementation. Figma also supports collaborative design and version control, allowing multiple team members to review and refine the system interface in real time. This ensures consistency in design and improves clarity when presenting the proposed system to stakeholders. Overall, Figma serves as an effective technology for illustrating the functional flow and user interface of the Campus Transport & Shuttle Management System.

## **2.5 Summary**

This project reviewed concepts related to Enterprise Architecture (EA) and its relevance to higher education institutions, with a particular focus on Universiti Tun Hussein Onn Malaysia (UTHM). This highlighted how universities rely on multiple enterprise systems to manage academic, administrative, and support services, and how poor system integration can lead to fragmented workflows, data silos, and inefficient decision-making.

The discussion on Enterprise Architecture explained its definition, core concepts, and four main architectural layers, which are Business, Application, Data, and Technology Architecture. EA was shown to be an important strategic tool that aligns IT systems with organizational goals, promotes interoperability, and supports scalability. Previous studies confirmed that EA plays a significant role in supporting smart campus

initiatives by enabling integrated services, real-time data sharing, and improved governance across university systems.

The related sub-systems focused on the proposed Campus Transport & Shuttle Management System. The system was designed to address limitations in UTHM's existing transport solution, which operates as a standalone system and lacks integration with other enterprise systems. The proposed system introduces features such as live shuttle tracking, ETA prediction, occupancy monitoring, user feedback, and analytics dashboards to support better service delivery and decision-making.

Finally, the technology used in the system is the use of Figma as a cloud-based UI/UX design and prototyping tool. Figma supports the visual design of system interfaces, interactive prototypes, and functional workflows, allowing the proposed Campus Transport & Shuttle Management System to be clearly illustrated and evaluated. The use of Figma enables collaborative design, consistency in user interface development, and effective communication of system functionality, while providing a clear blueprint for future implementation in alignment with Enterprise Architecture principles.

In conclusion, this project establishes a strong theoretical and technological foundation for the proposed enterprise-level Campus Transport & Shuttle Management System. It demonstrates how applying EA concepts and modern cloud-based technologies can help UTHM move toward a more integrated, efficient, and data-driven smart campus ecosystem.

## **CHAPTER 3 : METHODOLOGY**

### **3.1 Introduction - explain chosen methodology**

We have decided to choose Waterfall Methodology for this project. As known, Waterfall Methodology is a traditional model in which each phase of development must be completed and approved before moving on to the next. It ensures that project progress is well organized by understanding and following the objective of the project. Waterfall Methodology consists of phases such as requirement analysis, design, implementation, testing, deployment, and maintenance. This methodology was chosen for the project because the system requirements are well-defined and stable, making a linear and sequential approach suitable. Waterfall allows for comprehensive documentation and structured planning, which aligns with Enterprise Architecture principles and ensures clarity for both developers and stakeholders. By following Waterfall, the project can systematically address each aspect of the system such as from shuttle route display and live tracking to user feedback and analytics. This will reduce the risk of miscommunication and ensure that each module meets its intended objectives.

### **3.2 Chosen Methodology**

In the requirement phase, all functional and non-functional requirements of the system are gathered and documented. Functional requirements include the ability to display shuttle routes and schedules, track shuttles live, predict estimated arrival times (ETA), monitor shuttle occupancy, and collect user feedback and complaints. Non-functional requirements include system performance, reliability, usability, and security. The stakeholders, such as students and admin, are identified, and their needs are carefully analyzed. Clear documentation of these requirements ensures that the subsequent design phase can be accurately aligned with user expectations.

For the design phase, we focused on planning the architecture and workflows based on the documented requirements. The design specifies the structure and layout of user interfaces for students and admin, while defining the application logic for live shuttle tracking, ETA prediction, and shuttle occupancy monitoring. The design also

includes detailing how route schedules, tracking information, and user feedback will be stored, accessed, and integrated across the system. By clearly mapping the interactions between different modules and ensuring consistency in data flow and functionality, this phase provides a comprehensive blueprint that guides the development and implementation of the system.



### **3.3 Phases of Methodology**

#### **3.3.1 Phase 1 - Planning**

##### **Defining Project Scope**

During the planning phase, the project scope is carefully determined to ensure that the development process remains focused and achievable. Although the overall project involves designing an Enterprise Architecture that covers multiple operational areas within UTHM, the implementation phase concentrates solely on **Module 5: Campus Transport & Shuttle Management System**.

This module includes six main components, which are Route and Schedule Display, Live Shuttle Tracking, Estimated Time of Arrival (ETA) prediction, Shuttle Occupancy Monitoring, User Feedback, and Transport Analytics. Clearly defining the scope at an early stage helps prevent unnecessary feature expansion and ensures that the project objectives can be achieved within the given timeframe and academic requirements.

##### **Identifying Stakeholders**

Identifying stakeholders is an important activity in the planning phase, as it ensures that the system is designed according to actual user needs. For this project, two primary stakeholders are involved. The first group is **UTHM students**, who will use the system to access shuttle schedules, monitor shuttle locations in real time, check arrival estimates, and submit feedback or complaints regarding transport services. The second group is **UTHM administrators**, who are responsible for overseeing shuttle operations, monitoring system performance, managing feedback submissions, and analysing transport-related data through the system dashboard.

Focusing on these two stakeholder groups allows the system to balance usability for students with effective management and monitoring capabilities for administrators.

##### **Initial Feasibility Study**

An initial feasibility study is conducted to assess whether the proposed system can be successfully implemented. From a technical perspective, the use of Figma as a

prototyping tool provides a suitable environment for designing and validating user interfaces and system workflows. Figma supports rapid prototyping, collaborative design, and interactive mock-ups, enabling stakeholders to visualize system functionality early and verify system requirements.

From an operational standpoint, the system is considered feasible as it replaces existing manual practices, such as the use of Google Forms and Excel sheets, with a centralized digital solution. This transition improves efficiency, reduces redundant data entry, and enables faster access to transport information.

Overall, the Planning phase ensures that the project has a clear scope, well-defined stakeholders, and a realistic development approach. This structured preparation reduces potential risks and provides a strong base for the subsequent Analysis and Design phases of the Waterfall methodology.

### **3.3.2 Phase 2 - Analysis**

The Waterfall methodology outlines fairly distinct analysis tasks in this case, to define and document the functional requirements (FR) and nonfunctional requirements (NFR) of the proposed Campus Transport and Shuttle Management System. This is important in ensuring that the system is aligned with UTHM's enterprise architecture and smart campus initiatives and that it supports users' requirements from an operational perspective. In this way the requirements were derived from existing transport problems.

#### **Functional Requirements**

Functional requirements are statements of the system services or functions the system should provide. For example, the Campus Transport and Shuttle Management System shall process live tracking data so that it can present students and staff with a map showing the real-time locations of each shuttle. A seemingly simple feature, a real-time arrival tracker, reduces the agony and time spent at bus stops. With clocks coordinated, the system will calculate and air the ETA of each shuttle to bus stops. The shuttle app displays the occupancy level of the shuttle so students can determine if they are comfortable with the crowd level prior to boarding. Campus transportation that provides students with access to shuttle routes, schedules, and bus stop information. The system shall allow users to submit feedback or complaints related to shuttle services. Shuttle movement history data shall be stored. A clean analytics dashboard is essential to keep both users and administrators informed of shuttle performance, peak hours, etc. Ideally, the system would also allow administrators to control and manage shuttle routes and schedules and specify bus details. Usage and performance reports will be available to shuttle managers.

## **Non-Functional Requirements**

The section also contains non-functional requirements to ensure the overall quality of the system. The system will update the shuttle location and ETA information continuously and almost immediately. It should be something that students, staff and visitors to the university can easily access and understand. The system should run 24 hours a day, seven days a week, except for scheduled maintenance times. We wanted to make sure the system could support multiple shuttle buses and routes. And we didn't want the system to fail. Administrative features must be tightly controlled. The system design must be modular to allow future upgrades and maintenance without impacting the entire system. And students and staff will only be able to use mobile devices to reach the system via browsers (not for the administrative side).

### 3.3.3 Phase 3 - Design

In this design phase, we were able to translate the requirements into the working solution by building the enterprise architecture for the subsystem as well as the main UTHM system. For both the enterprise architecture, the mapping of the layers were conducted as follows:

**Business Layer** - This layer defined the business processes and stakeholder that interacted through the system to ensure the operational goals. The two primary stakeholders for the subsystem were the students that needed to access the real-time transport information and the admin staff of UTHM that analyzed the overall efficiency of the operations. We then mapped the overall workflow of the system each beginning from checking the route, tracking bus, boarding, and giving feedback for the student side while the admin side only supported monitoring analytics and resolving complaints.

**Application Layer** - This layer focused on designing the software components and how they interact to deliver functionality. We designed the interfaces of the system using the Figma as a prototype to demonstrate the actual functionalities by creating different views for each corresponding module including Route & Schedule Display, Live Shuttle Tracking, and User Feedback modules. Then we matched the modules to the processes involving it.

**Data Layer** - This data layer was designed to eliminate the silos of the system as well as ensuring data integrity. Through this layer, we created a data model containing Student and Staff Master data, route data, schedule, data, gps coordinate, shuttle occupancy data, speed tracking data, and list of complaint record data. Some of the data were integrated with the legacy system to authenticate users and link the feedback form to the specific users, Dedicated stores for Route, Schedule, GPS Coordinates, and Shuttle Occupancy to handle real-time tracking, and Storage for Complaint Records and Real-Time Speed data to support the analytics dashboard.

**Technology Layer** - The layer outlines the infrastructure that the system runs on including the notifications that alert the student about the bus arrival, cloud database that will host all of the data available, api layer to connect the system with the data through fetching, and UTHM Single Sign-On to secure access and link to student and staff profile.

### 3.4 Project Planning Schedule

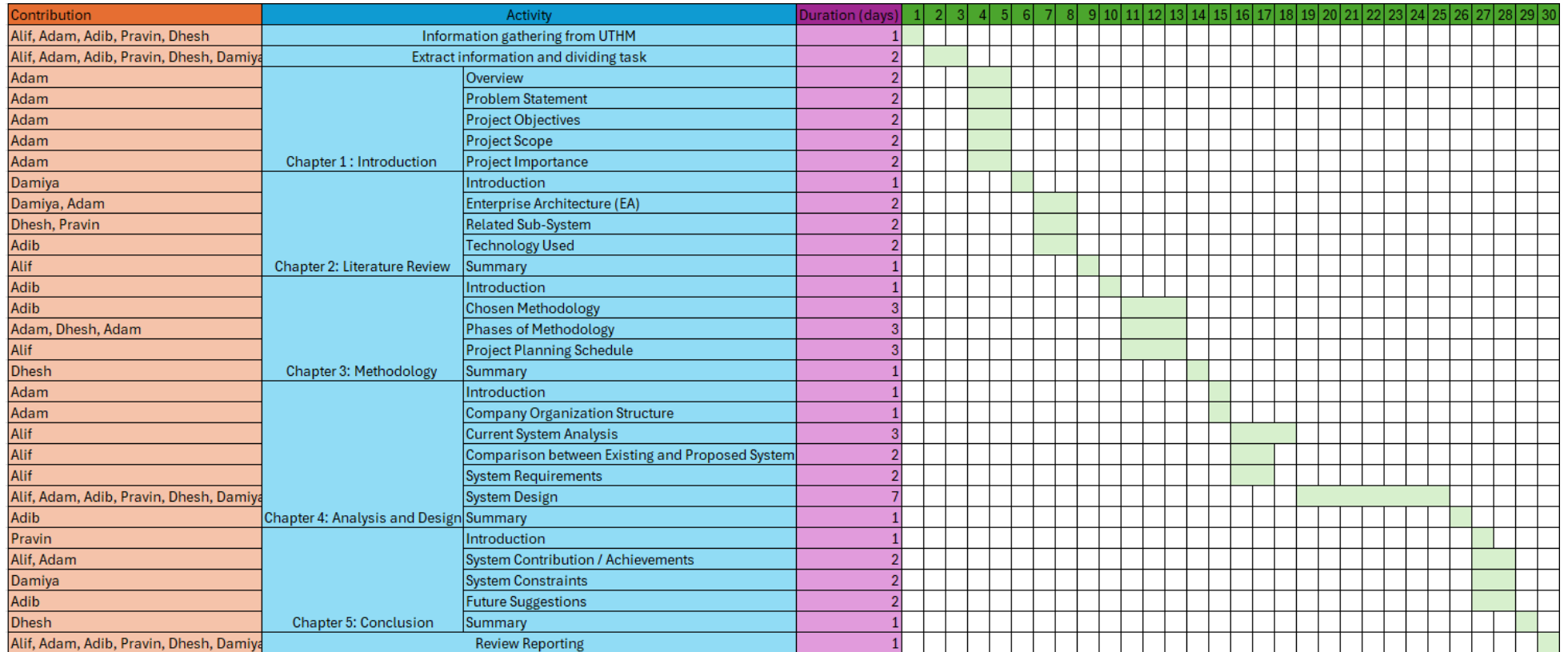


Figure 3.1 Project Planning Schedule (Gantt Chart)

### **3.5 Summary**

This chapter explains the methodology used to develop the proposed Campus Transport and Shuttle Management System for Universiti Tun Hussein Onn Malaysia (UTHM). The Waterfall methodology was selected because of its structured and sequential approach, making it suitable for a project with well-defined requirements. By following this methodology, the development process is organized into clear phases with proper documentation, ensuring alignment with Enterprise Architecture principles and reducing the risk of miscommunication.

The chapter focused on the Planning, Analysis, and Design phases. The Planning phase defined the project scope and identified UTHM students and administrators as the main stakeholders, while confirming the feasibility of using Figma as a prototyping tool to design and validate the system interface and workflows. The Analysis phase outlined the functional and non-functional requirements of the system, followed by the Design phase, which mapped the Business, Application, Data, and Technology layers into a structured enterprise architecture. Overall, this methodology provides a solid foundation for developing an integrated and efficient campus transport system that supports UTHM's smart campus goals.



## 4.2 Company Organization Structure

university's digital systems, such as the student database and the mobile app. Second, the Chief Operating Officer (COO) manages the Facility Management Office, which is responsible for physical assets like buses and bus stops. For the new transport system to work effectively, the IT team must work closely with the facility managers to connect the digital app with the actual bus operations.

### **4.3 Current System Analysis**

Currently, UTHM offers shuttle bus monitoring via two main platforms, which include the MyUTHM mobile application and real-time shuttle tracking via the UTHM/Katsana system. A web-based shuttle tracking system that is connected to the Katsana GPS platform is used by UTHM. Through the UTHM transportation website, this technology enables students to see the shuttle buses' current locations on a map. Although it only focuses on position tracking, it assists users in knowing the shuttle's current location. Additionally, the MyUTHM mobile app offers live shuttle tracking, making it easy for students to use phones to keep an eye on shuttle progress. However, the app primarily shows real-time location without any further operational information, similar to the web system.

The current system has a number of drawbacks. These include the lack of shuttle occupancy monitoring to show the number of passengers, lack of Estimated Time of Arrival (ETA) information for each bus stop, the lack of analytics or performance monitoring tools for management purposes, and the lack of a structured method for gathering and evaluating user complaints or feedback. The efficiency of shuttle planning, service optimization, and decision-making is reduced by these limitations.

#### 4.4 Comparison between Existing and Proposed System

The table below shows the difference of capabilities between the current existing system of the Katsana UTHM and our Campus and Shuttle Management subsystem in tabular form.

| Features                                      | Existing      | Proposed  |
|-----------------------------------------------|---------------|-----------|
| Live Shuttle Tracking                         | Available     | Available |
| Shuttle Occupancy Monitoring                  | Not Available | Available |
| Estimated Time of Arrival (ETA) for Each Stop | Not Available | Available |
| Analytics & Performance Monitoring            | Not Available | Available |
| Student Feedback & Complaint Analysis         | Not Available | Available |

#### 4.5 System Requirements

System requirements define the basis on which software will function. They include a wide range of topics, including outputs, inputs, processes, performance, and controls.

##### 4.5.1 Functional Requirements

- The system shall display real-time shuttle location on a map for students and staff.
- The system shall calculate and display Estimated Time of Arrival (ETA) for each shuttle bus stop.
- The system shall monitor and display shuttle occupancy level.
- The system shall allow students to view shuttle routes and bus stop information.
- The system shall allow users to submit feedback or complaints related to shuttle services.
- The system shall store shuttle movement history data for analysis purposes.

- The system shall provide an analytics dashboard for administrators to monitor shuttle performance.
- The system shall allow administrators to manage shuttle routes, schedules and bus details.
- The system shall generate reports based on shuttle usage and performance data.

#### **4.5.2 Non-Functional Requirements**

- The system shall update shuttle location and ETA information in near real time.
- The system shall have a simple and user-friendly interface that can be easily used by students and staff.
- The system shall be available 24/7, except during scheduled maintenance periods.
- The system shall be able to support multiple shuttle buses and routes without performance degradation.
- The system shall provide accurate tracking and ETA calculations with minimal system downtime.
- The system shall restrict access to administrative features using user authentication and authorization.
- The system shall be designed in a modular manner to allow easy updates and maintenance.
- The system shall be accessible through mobile devices for users and web browsers for admin features.

## 4.6 System Design

### 4.6.1 Enterprise Architecture

#### UTHM Enterprise Architecture

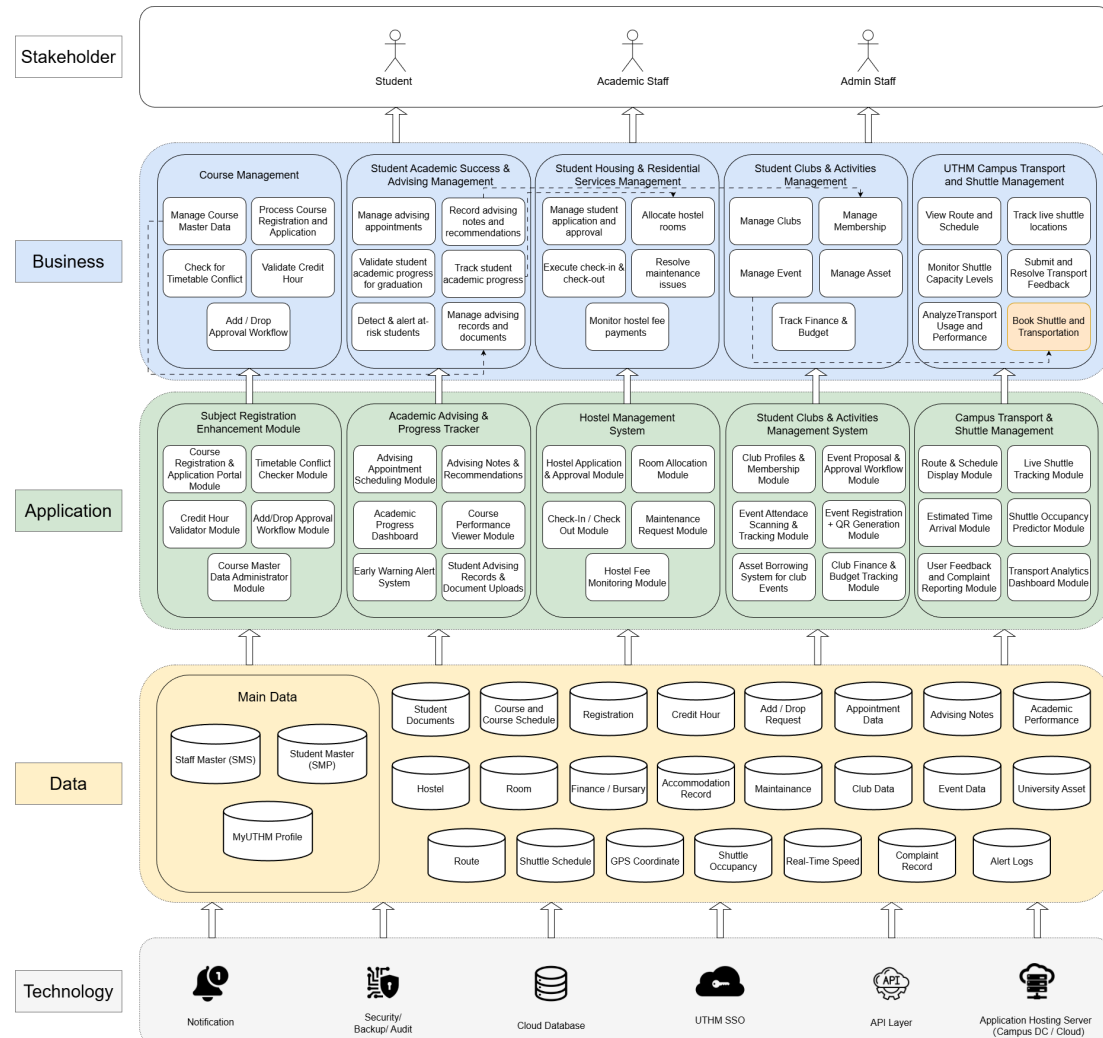


Figure 4.2 UTHM Enterprise Architecture Layers

**Business layer** - This layer represented the operational processes integrated inside the UTHM which include Course Management, Academic Success & Advising, Housing & Residential Services, Clubs & Activities, and Campus Transport. These segmentation divides the critical aspect of the administration in UTHM into several business processes.

Application layer - This application layer focuses on the software components that will display all the necessary interfaces to deliver the functionalities from the business layer. Therefore, it acts the bridge connecting the user needs and the data transferred from the data layer, making it visually represented as prototypes that demonstrates the functionality of the system.

Data layer - The Data Layer is designed to guarantee information integrity throughout the university's digital environment and remove data silos. It divides data into two primary categories: dynamic data, which contains real-time variables like shuttle occupancy, GPS locations, and vehicle speed, and static data, which includes things like set timetables and route specifics. Importantly, this layer links feedback or usage records to particular user profiles and verifies user identities by integrating with UTHM's legacy "Master Data" systems, namely the Student Master (SMP) and Staff Master (SMS). The design facilitates real-time analytics and customized user experiences by centralizing this data in cloud databases and allocating storage for storing logs and complaints.

Technology layer - This layer explain the infrastructure and technical services required to support the application including the data and software interfaces. It includes cloud database, api layer to connect and fetch data between system, as well as UTHM Single-Sign-On (SSO) gateway as a secure authentication for students and staff.

## Campus Transport & Shuttle Management Architecture

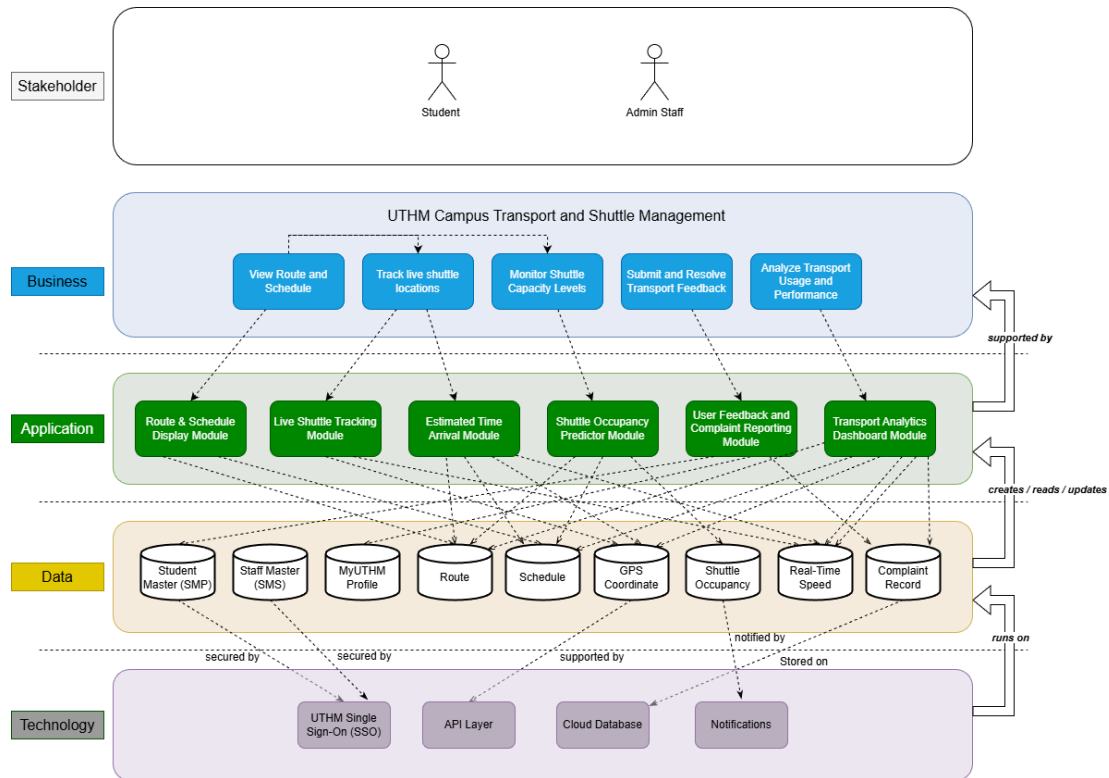


Figure 4.3 Campus Transport & Shuttle Management Architecture

Business & application layer - This layer categorizes the university's business processes across several departments that manage the campus transport and shuttle management, which includes viewing route schedules, tracking live shuttle locations, monitoring shuttle capacity level, submitting and resolving transport feedback, and analyzing transport usage performance. It links directly to the module that relates to and serves these business processes which consists of specialized modules like the Live Shuttle Tracking Module, Shuttle Occupancy Predictor Module, and the Transport Analytics Dashboard Module. This direct mapping ensures that every software feature serves a clear operational purpose.

Data layer - This layer contained all of the static data like Schedule and route, and dynamic data such as shuttle occupancy, gps coordinate, as well as real-time speed. All of these data were also integrated with the institutional Student Master (SMP), MyUTHM Profile, and Staff Master (SMS) databases. This integration allows the system to verify user identities and potentially personalize the viewing experience based on the user's profile.

Technology layer - The technology layer is designed to support the data and processes required for these subsystems, such as GPS simulation and live tracking. The security is also implemented via UTHM SSO, and notifications are employed to alert users about arrivals or service disruptions, ensuring a responsive and secure user experience.



#### 4.6.2 System Architecture

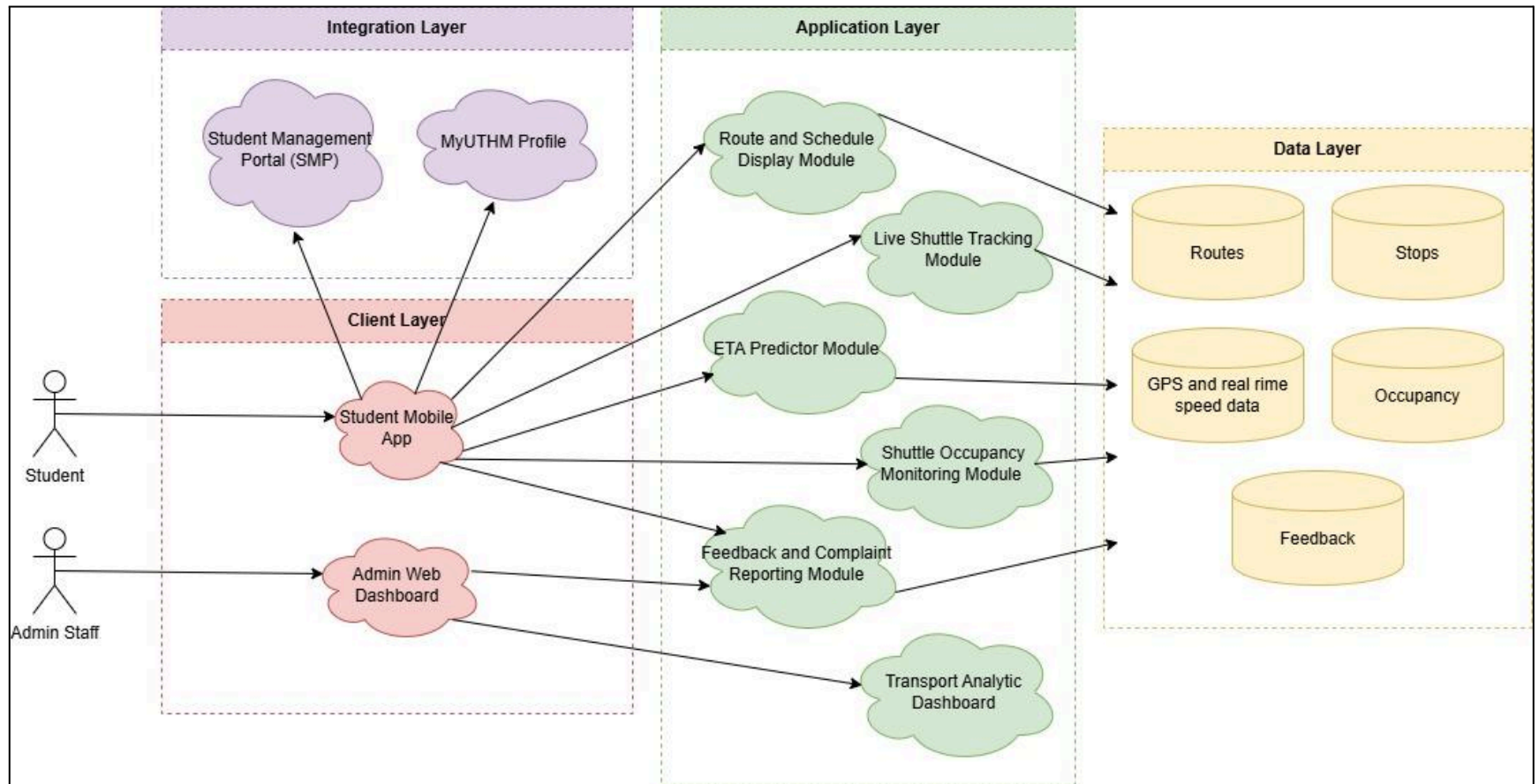


Figure 4.4 System Architecture Diagram

### 4.6.3 Project Design Tools

#### 4.6.3.1 Use Case Diagram

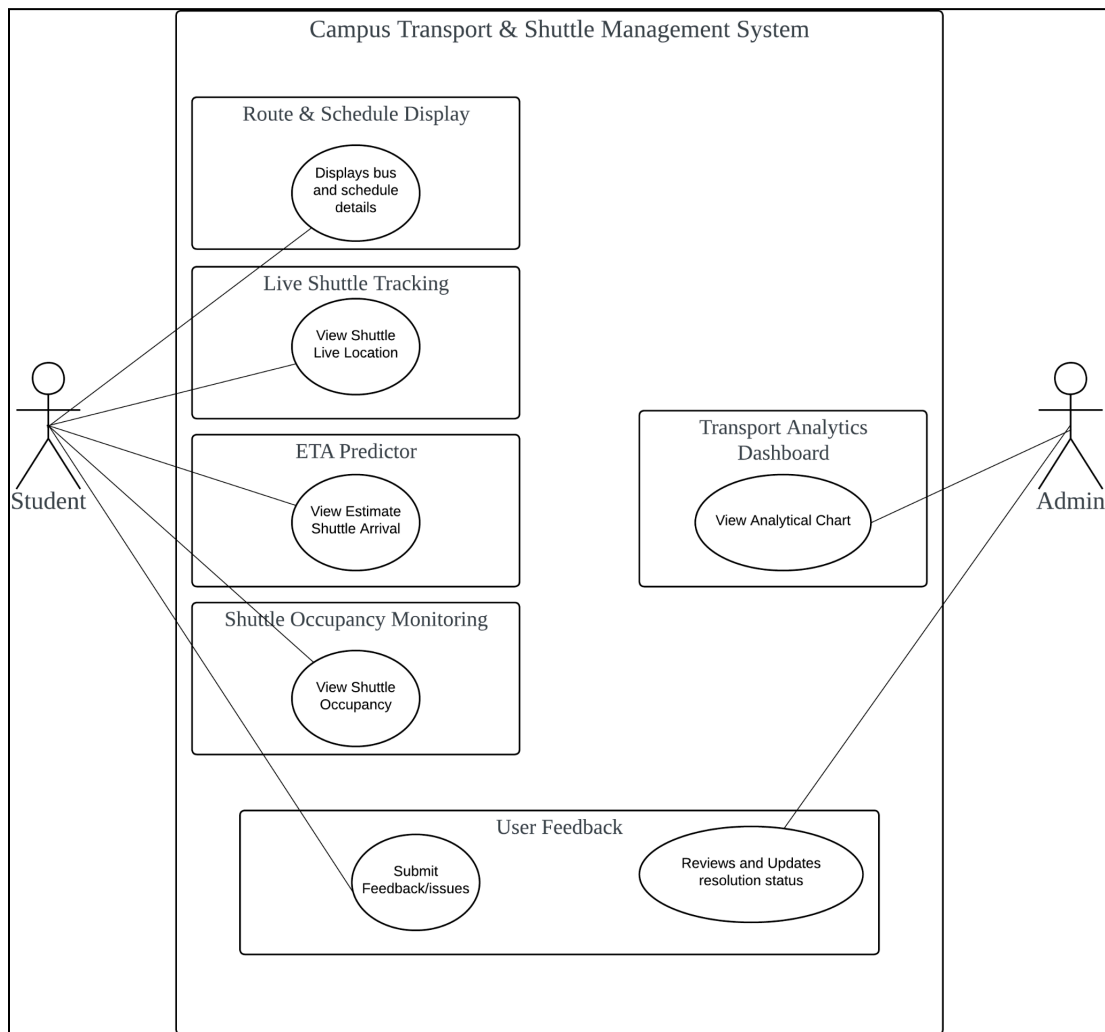


Figure 4.5 Use Case Diagram: Campus Transport & Shuttle Management System

#### 4.6.3.2 Use Case Description

##### Route & Schedule Display Module

| Use Case Description: Route and Schedule Display                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors:</b><br>Students                                                                                                                                                                                                                                                                                         |
| <b>Preconditions:</b><br>User has accessed the Campus Transport & Shuttle Management System                                                                                                                                                                                                                        |
| <b>Flow of Events:</b> <ol style="list-style-type: none"><li>1. Users access the Campus Transport &amp; Shuttle Management System.</li><li>2. The users navigate to the Route and Schedule Display module.</li><li>3. The system displays available shuttle routes.</li><li>4. The user selects a route.</li></ol> |
| <b>Postconditions:</b><br>Users are able to view the shuttle routes and schedules for campus transportation.                                                                                                                                                                                                       |

## Live Shuttle Tracking Module

| Use Case Description : Live Shuttle Tracking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors:</b><br>Student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description:</b><br>Allows the user to view the real-time position of campus shuttle buses on a map interface to track their movement across the UTHM campus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Preconditions:</b><br>1. The user has logged in via the UTHM Single Sign-On (SSO).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Flow of Events:</b><br>1. The student navigates to the "Transport & Shuttle" subsystem.<br><br>2. The system requests the current list of active shuttles from the database.<br><br>3. The student selects the "Shuttle Live Tracking" feature.<br><br>4. The student chooses the bus to view for its location from the list of shuttle buses available.<br><br>5. The system requests the GPS Simulation Service to fetch the current coordinates of the shuttle chosen.<br><br>6. The system displays the entire route of the shuttle in a single line from start to end with the bus icon on it to indicate the current shuttle location.<br><br>7. The system periodically refreshes the coordinates (variable updates) to animate the shuttle's movement. |
| <b>Postconditions:</b><br>The user sees the current location of the selected shuttle updated on the map in live.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Alternative Flows:</b><br>A1. No Shuttles Active: If any shuttles are currently not operating, they will not be listed on the shuttle lists.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Estimated Arrival Time (ETA) Predictor Module

| User story: View Shuttle ETA                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Story Description</b><br>As a student, I want to view the estimated arrival time (ETA) of the campus shuttle so that I can plan my time and avoid waiting too long at the bus stop.                                                                                                                                                                  |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The student opens the ETA Predictor system.</li><li>2. The student selects a bus stop.</li><li>3. The system retrieves the current shuttle location.</li><li>4. The system calculates the estimated arrival time.</li><li>5. The system displays the ETA countdown to the student.</li></ol> |
| <b>Alternative flow 1:</b><br>If shuttle location data is unavailable, the system displays a message indicating that the ETA cannot be calculated.                                                                                                                                                                                                           |
| <b>Alternative flow 2:</b><br>If the student does not select a bus stop, the system prompts the student to select one before proceeding.                                                                                                                                                                                                                     |
| <b>Precondition:</b><br>The student has access to the ETA Predictor system.                                                                                                                                                                                                                                                                                  |
| <b>Postcondition:</b><br>The estimated arrival time of the shuttle is displayed to the student.                                                                                                                                                                                                                                                              |
| <b>Exception Flow</b><br>System error occurs while retrieving shuttle data, and the system displays an error message.                                                                                                                                                                                                                                        |

## Shuttle Occupancy Monitoring Module

| User Story: View Shuttle Occupancy Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Story Description</b><br>As a student, I want to view the shuttle occupancy level so that I can decide whether to board the shuttle or wait for a less crowded one.                                                                                                                                                                                                                                                                                                                                                      |
| <b>Flow of Events</b> <ol style="list-style-type: none"><li>1. The student opens the Shuttle Occupancy Monitoring module.</li><li>2. The system displays a list of available shuttle routes.</li><li>3. The student selects a shuttle route.</li><li>4. The system retrieves the current simulated occupancy data for the selected shuttle.</li><li>5. The system determines the occupancy level (Low / Medium / High).</li><li>6. The system displays the shuttle occupancy level and related details to the student.</li></ol> |
| <b>Alternative Flow</b><br>If the student does not select a shuttle route, the system prompts the student to select one before displaying occupancy information.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Precondition</b><br>The student has access to the Shuttle Occupancy Monitoring module through the campus transport system.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Postcondition</b><br>The shuttle occupancy level is successfully displayed to the student to support journey planning.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Exception Flow</b><br>If a system error occurs while retrieving occupancy data, the system displays an error message and advises the student to try again later.                                                                                                                                                                                                                                                                                                                                                              |

## User Feedback & Complaint Reporting

| User Story: User Feedback & Complaint Reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>User Story Description</b><br>As a student, I want to submit feedback or complaints regarding campus shuttle services so that service issues can be reported and improved efficiently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Flow of Events</b> <ol style="list-style-type: none"><li>1. The student logs into the system using UTHM Single Sign-On (SSO).</li><li>2. The student navigates to the Campus Transport &amp; Shuttle Management System.</li><li>3. The student selects the User Feedback &amp; Complaint Reporting module.</li><li>4. The system displays the feedback and complaint submission form.</li><li>5. The student selects a complaint or feedback category.</li><li>6. The student enters the complaint or feedback description.</li><li>7. The student submits the feedback.</li><li>8. The system validates the input.</li><li>9. The system stores the feedback or complaint in the database.</li><li>10. The system displays a submission confirmation message to the student.</li></ol> |
| <b>Alternative Flow</b><br>If the student submits the form without completing required fields, the system prompts the student to complete the missing information before submission.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Precondition</b><br>The student has successfully logged in via the UTHM Single Sign-On (SSO) and has access to the Campus Transport & Shuttle Management System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Postcondition</b><br>The feedback or complaint is successfully recorded in the system and is available for administrator review and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Exception Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

If a system error occurs while submitting the feedback, the system displays an error message and advises the student to try again later.



## Transport Analytics Dashboard

| User Story: Analyze Transport Usage and Performance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>User Story Description</b></p> <p>As an admin staff member, I want to analyze campus transport usage and performance so that I can monitor operations, evaluate efficiency, and support informed decision-making.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Flow of Events</b></p> <ol style="list-style-type: none"><li>1. The admin logs into the system using the UTHM Single Sign-On (SSO).</li><li>2. The admin navigates to the Transport &amp; Shuttle subsystem.</li><li>3. The system requests transport-related data from the database.</li><li>4. The system checks the availability of transport data.</li><li>5. The system loads the transport analytics dashboard.</li><li>6. The admin selects analysis criteria such as date, route, and time</li><li>7. The system retrieves the required data from the database.</li><li>8. The system processes transport usage and performance metrics.</li><li>9. The system displays analytics charts and KPI summaries to the admin.</li></ol> |
| <p><b>Alternative Flow</b></p> <p>If transport data is not available, the system displays a <b>No Data Available</b> message on the analytics dashboard.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Precondition</b></p> <p>The admin staff has successfully logged in via the UTHM Single Sign-On (SSO).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Postcondition</b></p> <p>Transport usage statistics, performance metrics, and analytical charts are displayed on the dashboard for admin monitoring and evaluation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Exception Flow**

If a system error occurs while retrieving or processing transport data, the system displays an error message and prompts the admin to try again later.

#### 4.6.3.3 Activity Diagram

##### Route & Schedule Display Module

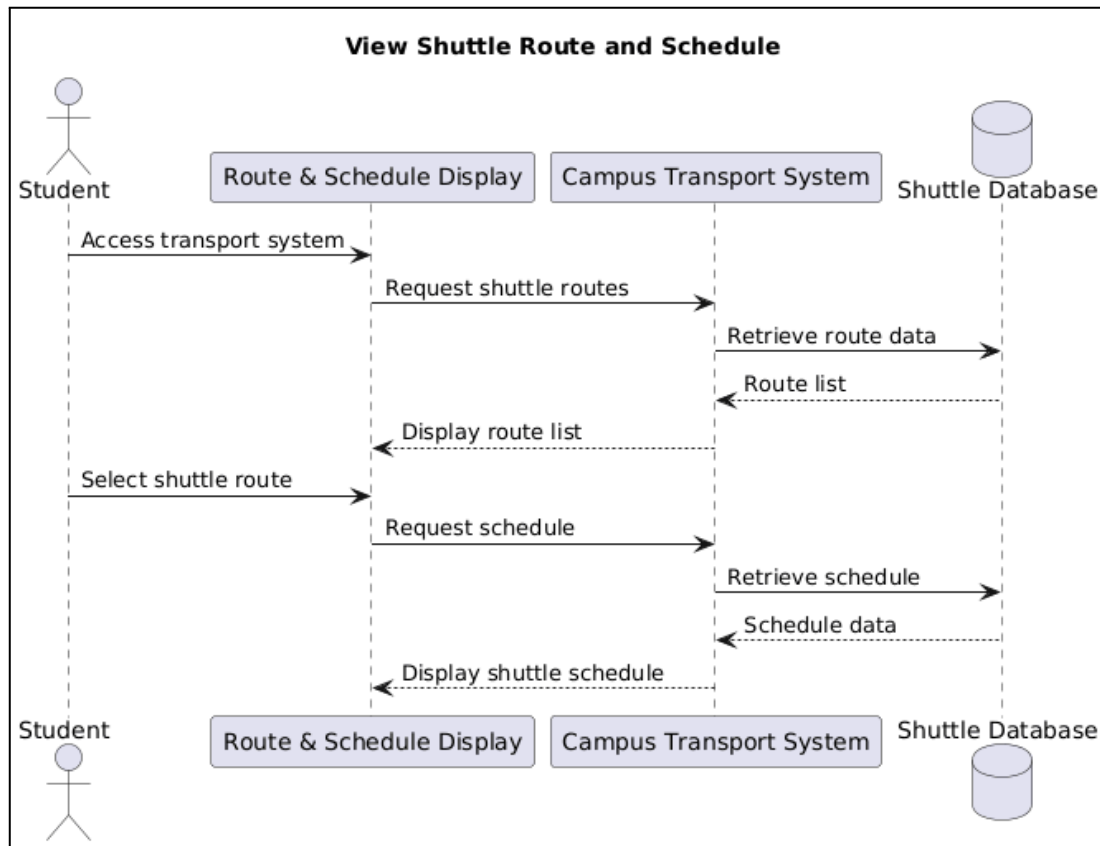


Figure 4.6 Activity Diagram: View Shuttle Route and Schedule

## Live Shuttle Tracking Module

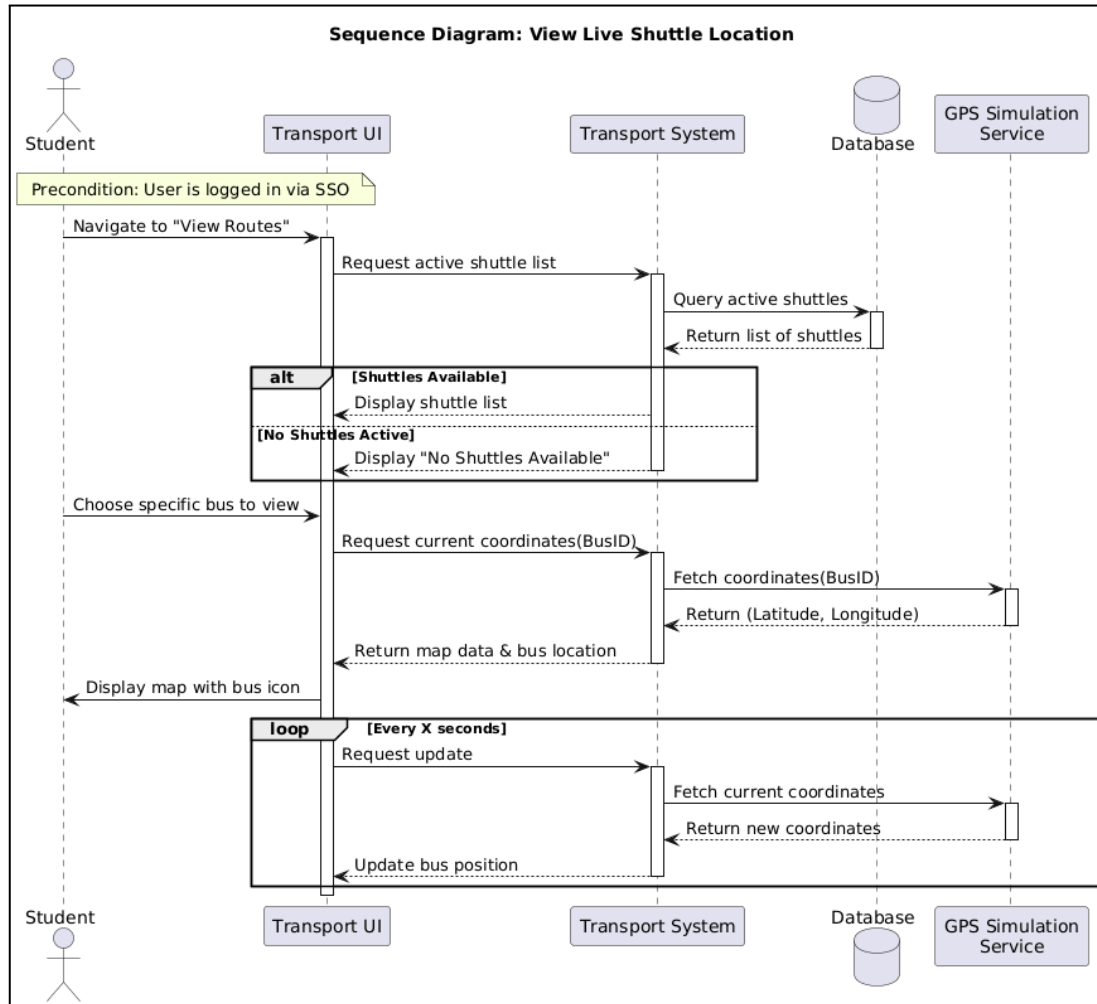


Figure 4.7 Activity Diagram: View Live Shuttle Location

## Estimated Arrival Time (ETA) Predictor Module

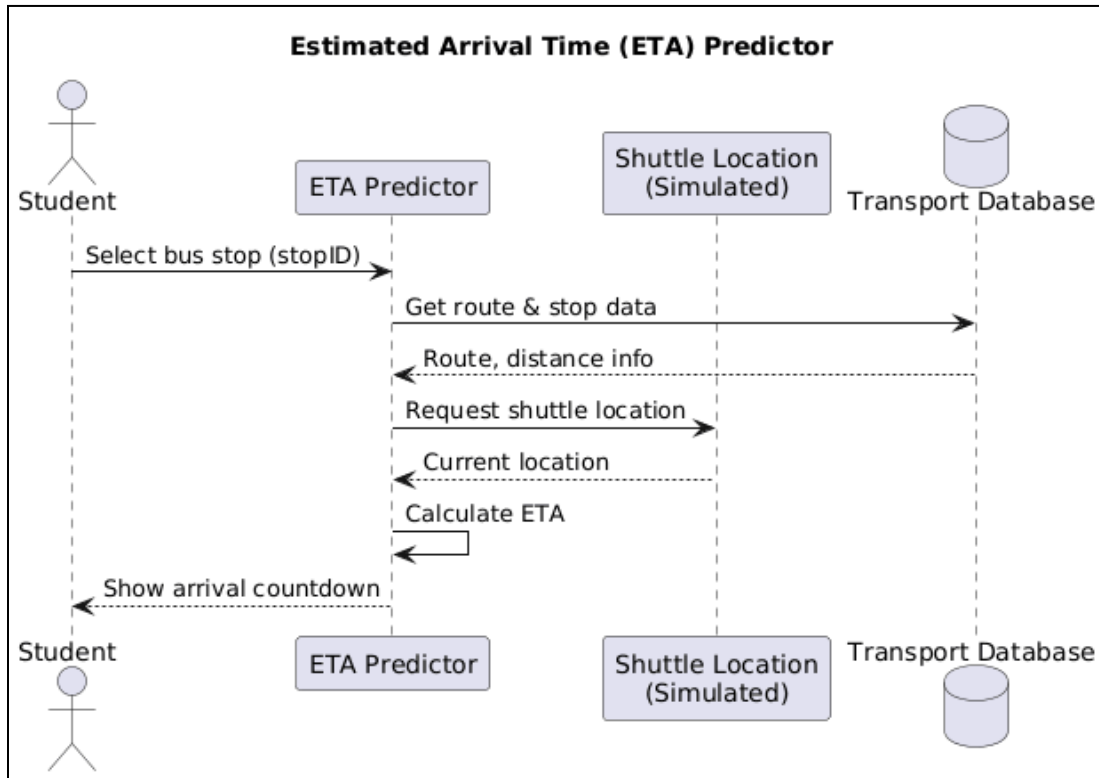


Figure 4.8 Activity Diagram: ETA Predictor Module

## Shuttle Occupancy Monitoring Module

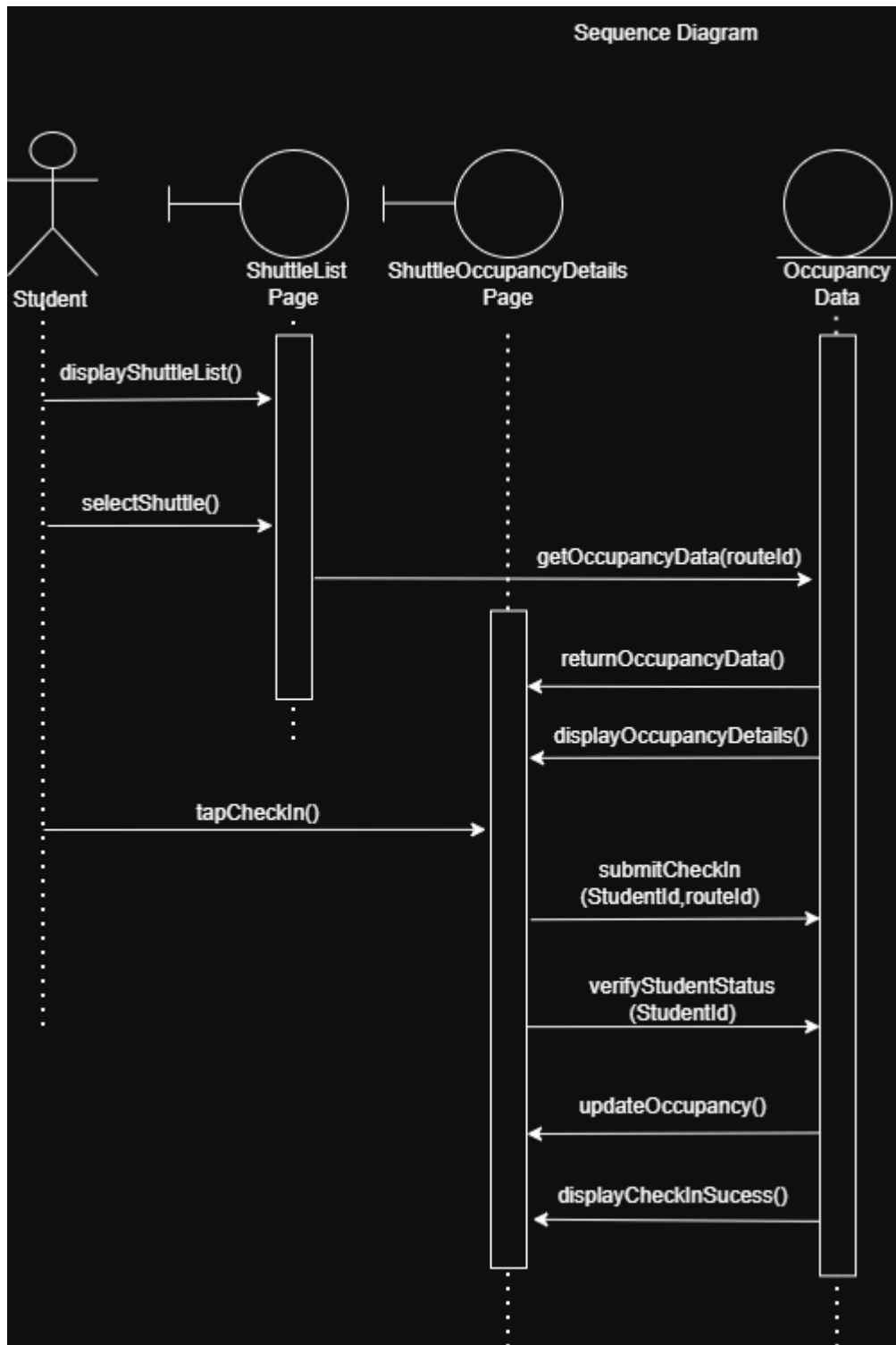


Figure 4.9 Activity Diagram: Shuttle Occupancy Monitoring

## User Feedback & Complaint Reporting

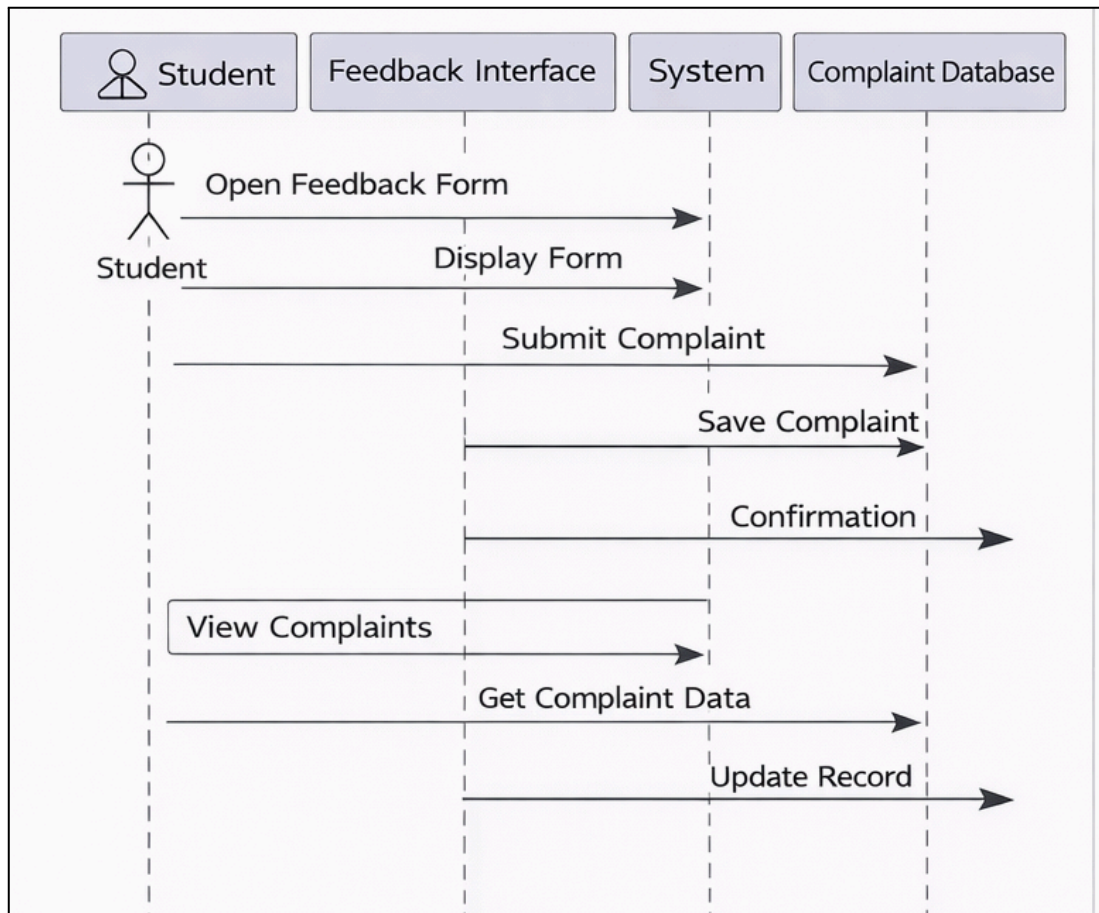


Figure 4.10 Activity Diagram: User Feedback & Complaint Reporting

## Transport Analytics Dashboard Module

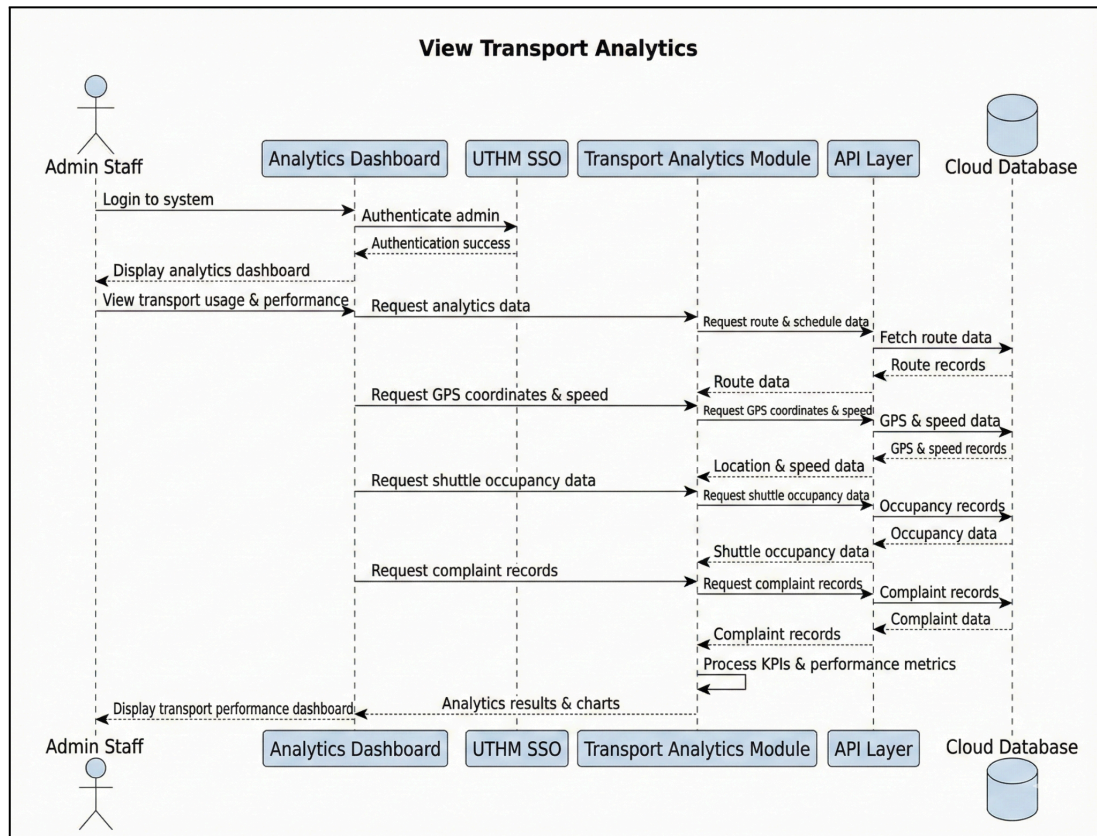


Figure 4.11 Activity Diagram: Transport Analytics Dashboard



#### 4.6.3.4 Sequence Diagram

##### Route & Schedule Display Module

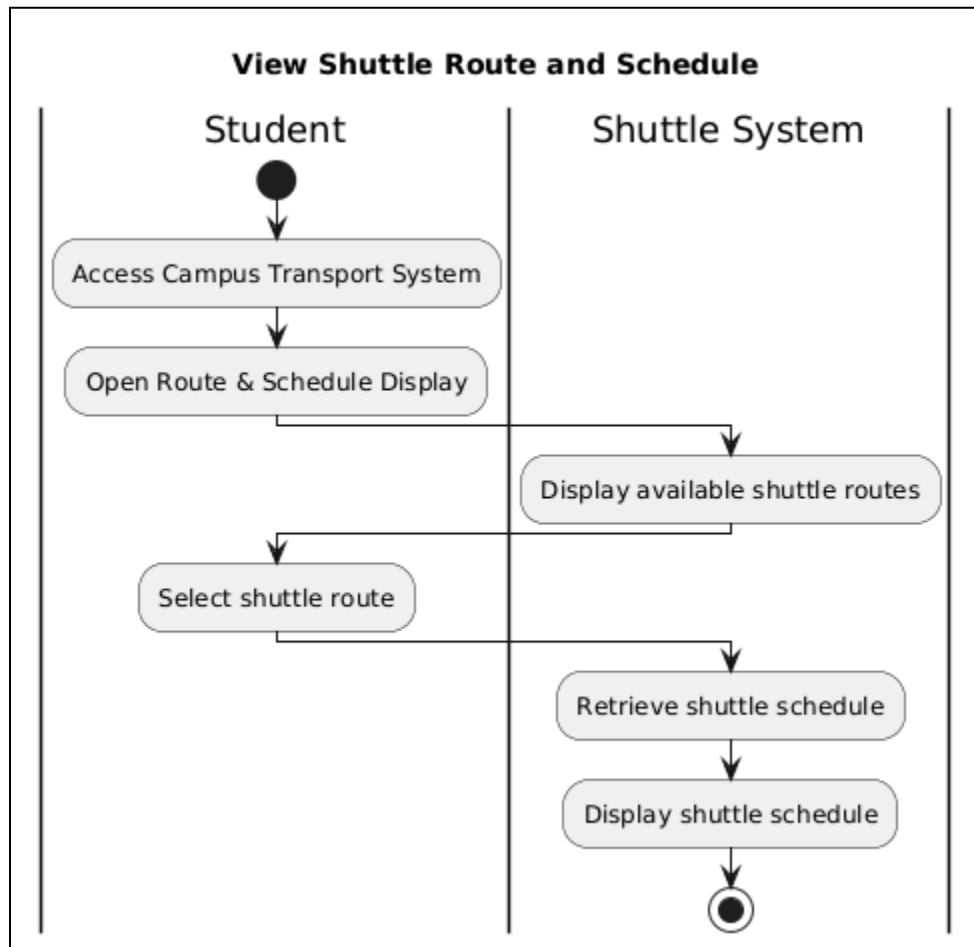


Figure 4.12 Sequence Diagram: View Shuttle Route and Schedule

## Live Shuttle Tracking Module

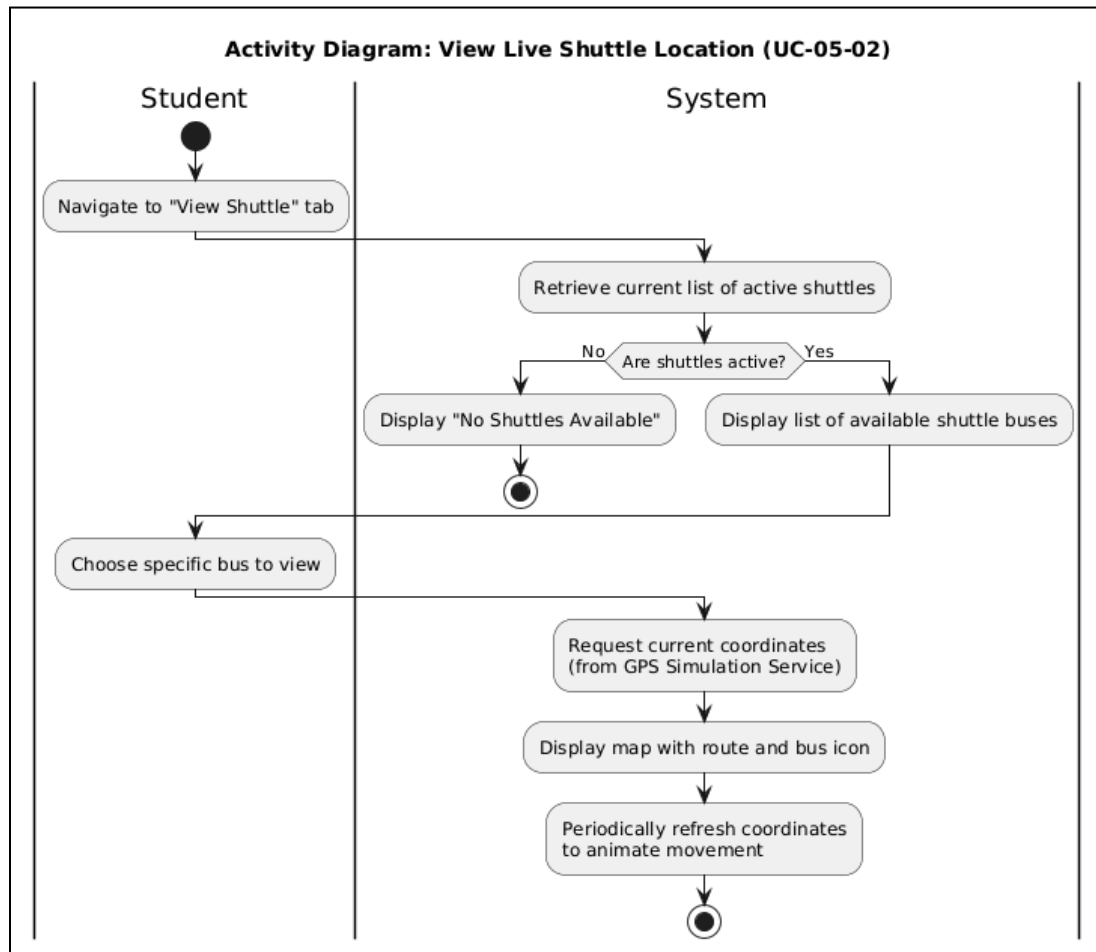


Figure 4.13 Sequence Diagram: View Live Shuttle Location

## Estimated Arrival Time (ETA) Predictor Module

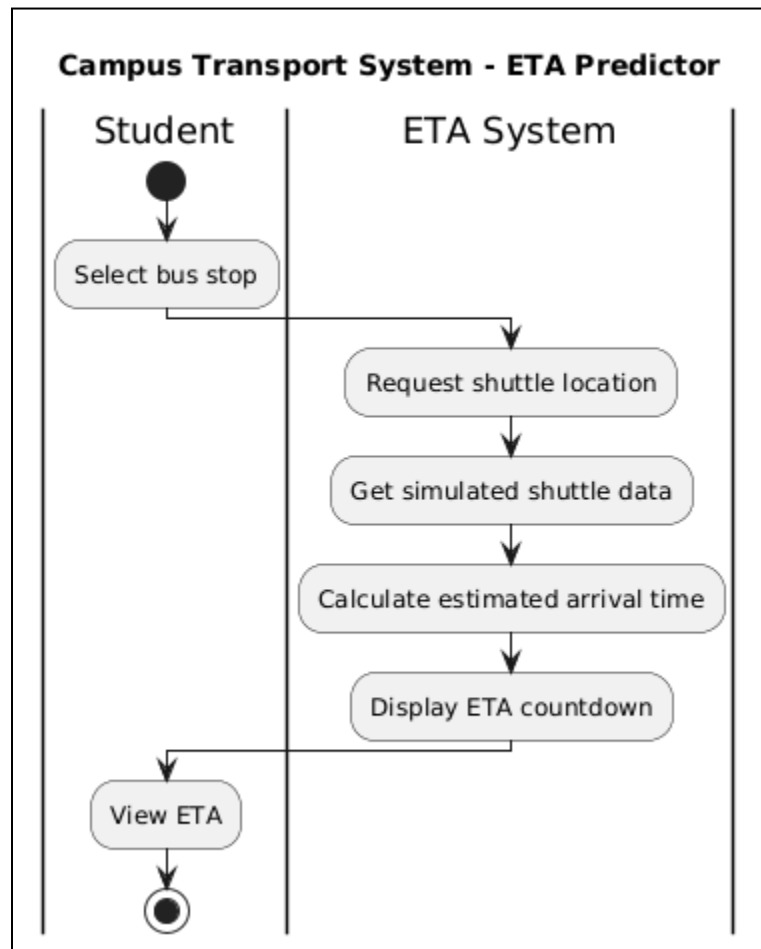


Figure 4.14 Sequence Diagram: Estimated Arrival Time (ETA) Predictor

## Shuttle Occupancy Monitoring Module

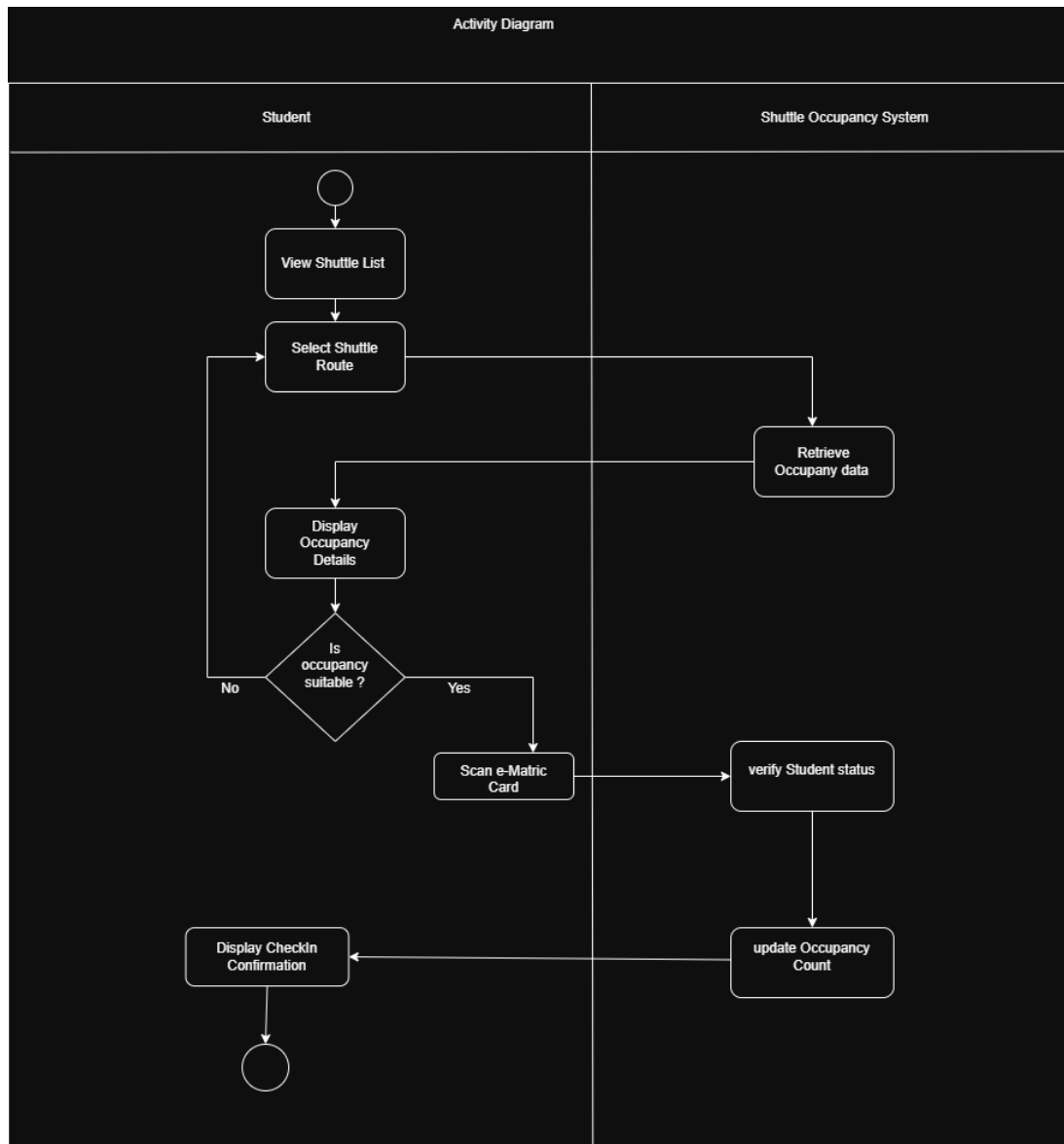


Figure 4.15 Sequence Diagram: Shuttle Occupancy Monitoring

## User Feedback & Complaint Reporting

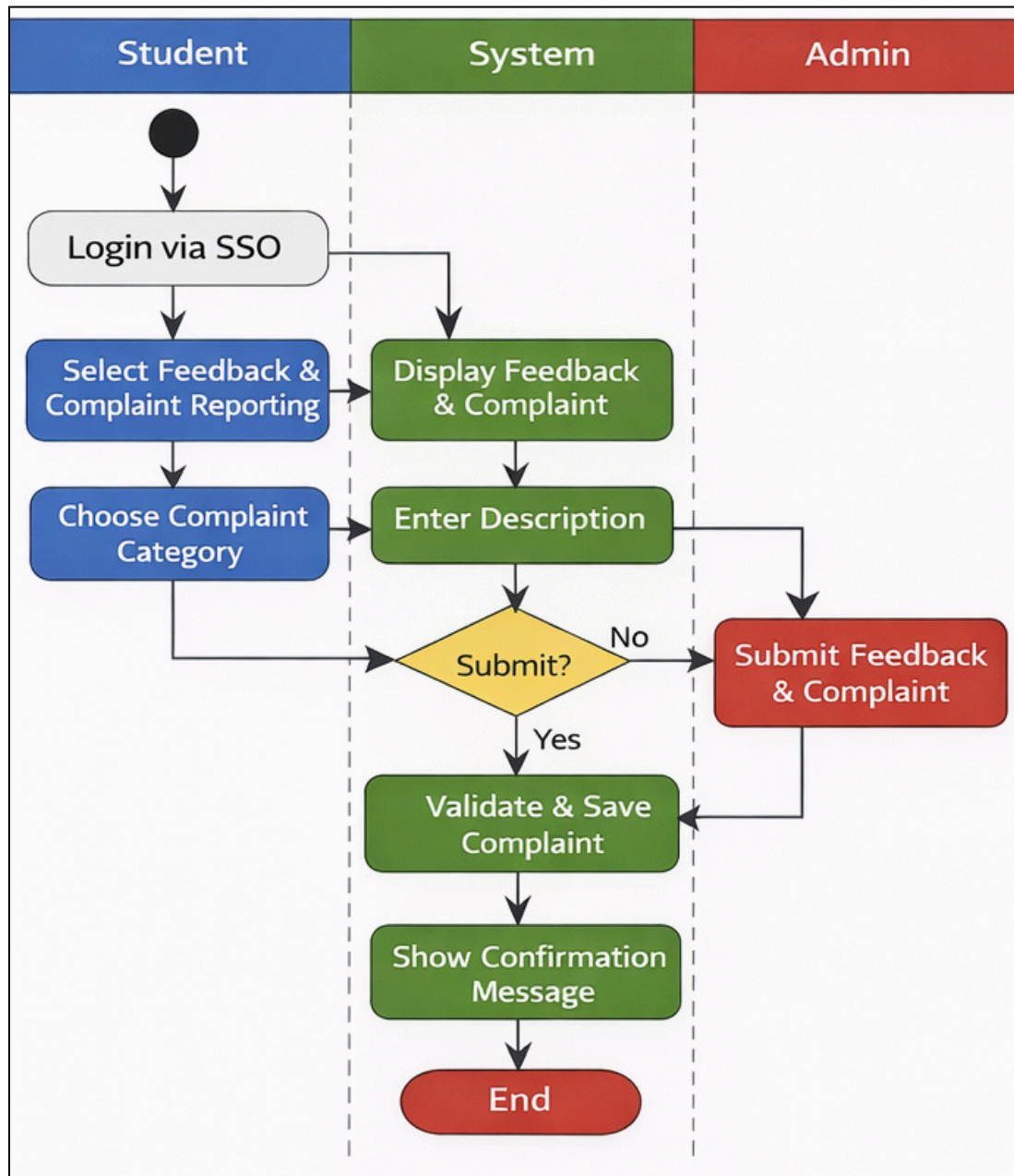


Figure 4.16 Sequence Diagram: User Feedback & Complaint Reporting

## Transport Analytics Dashboard Module

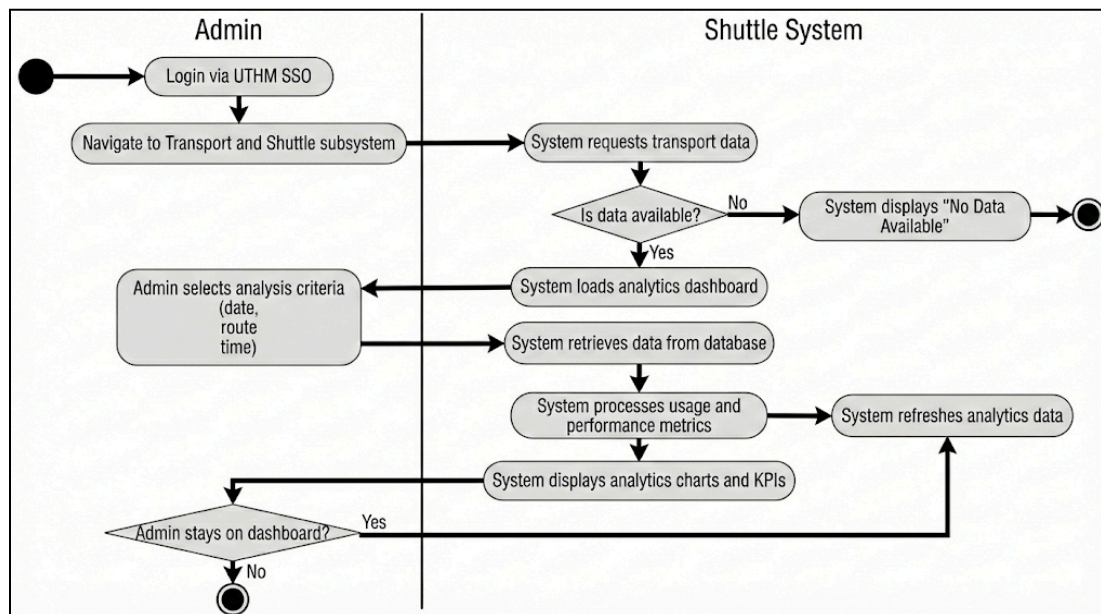


Figure 4.17 Sequence Diagram: Transport Analytics Dashboard

#### 4.6.3.5 Database Design – ER Diagram/Class Diagram/Normalization/Star Schema

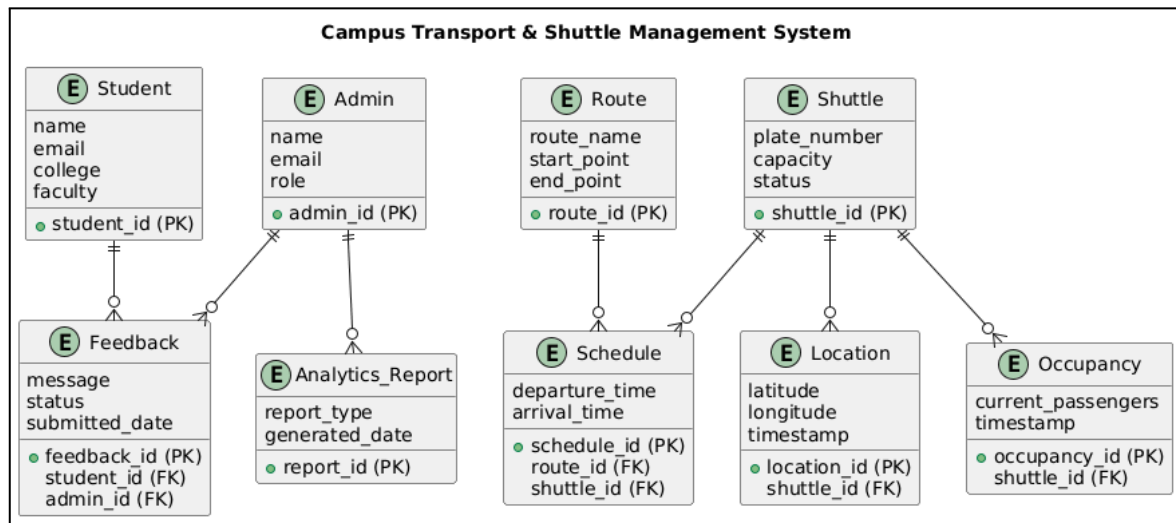


Figure 4.18 Entity Relationship (ER) Diagram

Based on the Database Design diagram in 4.6.3.5, user management controlled the data of the user that includes Student and Admin entities, which store essential information including name, email, faculty, college and role. These entities were directly linked to the Feedback entities that record feedback retrieved from the Student and passes it to admin to manage the complaint and give suggestion as well as directly aggregating into the analytic reports.

On the other hand, the core operational for the transport and shuttle management is owned by the Route, Schedule, and Shuttle entities. The Route entity defines the path for each unique shuttle from the Shuttle entities based on the plate number and bus code. The connection between the Route and Shuttle entities by the Schedule entity which uses foreign keys to assign specific shuttles to routes and sets their departure and arrival times, ensuring the system can accurately display static timetables .

To support the “Smart Campus” features, the database structure also include dynamic entities such as Location and Occupancy to support real-time tracking and analytics. This high-frequency data is likely aggregated into the Analytics\_Report entity, which

stores generated reports to help administrators analyze long-term performance trends and transport usage efficiency .



#### 4.6.3.6 Interface Design (individual)

##### Route & Schedule Display Module

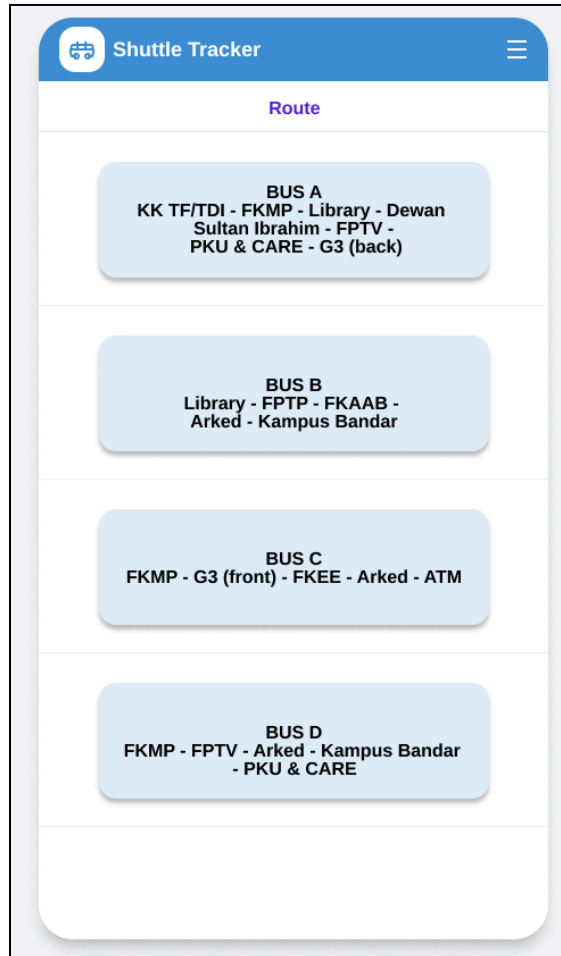


Figure 4.19 User Interface Design: Route & Schedule Display (Shuttle Tracker)

## Live Shuttle Tracking Module



Figure 4.20 User Interface Design: Live Shuttle Tracking Details

This page displays how the system track the location of the shuttle available inside the uthm for better navigation. The red bus icon shows the position of the bus prior to the current position of the user.

## Estimated Arrival Time (ETA) Predictor Module

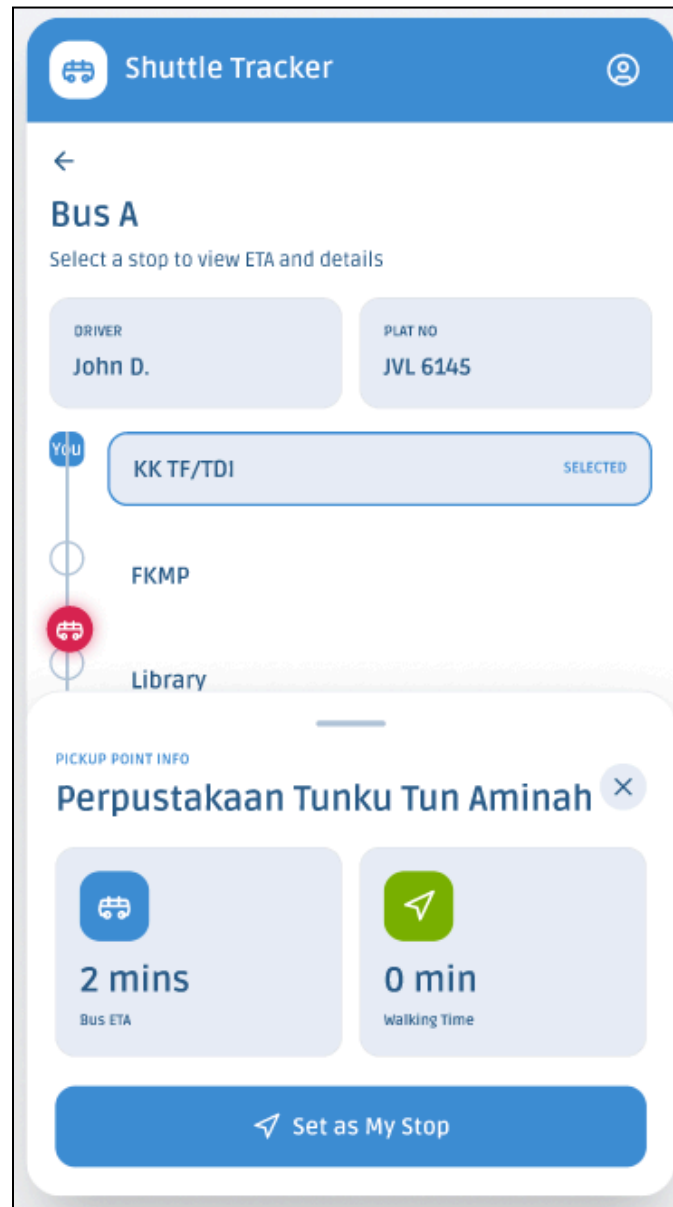


Figure 4.21 User Interface Design: ETA Predictor View

This page displays the ETA Predictor as a bottom pop-up. After the user selects a route and views the live shuttle location, they can choose a stop to see the estimated time for the bus to arrive at that stop. Users may also set the selected stop as “My Stop” to receive a notification when the bus arrives.

## Shuttle Occupancy Monitoring Module

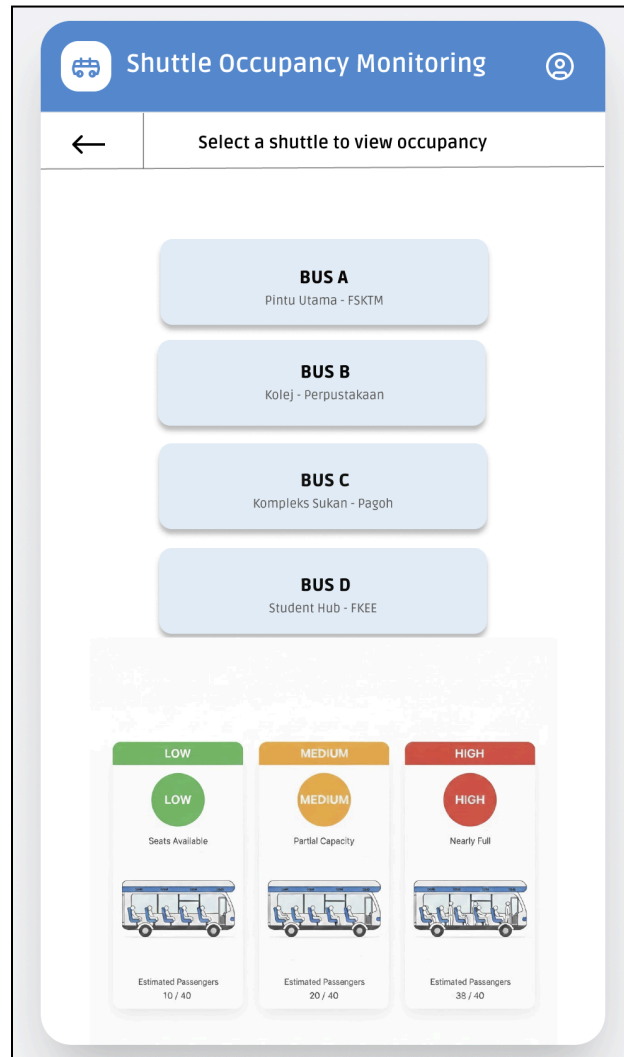


Figure 4.22 User Interface Design: Shuttle Occupancy List Overview

### Page 1: Shuttle List Overview Page

It lists campus shuttle routes and their current occupancy. The occupancy is measured in simple color codes which are (Low, Medium, High.) With the app, students can act as a real-time shuttle buddy, helping them evaluate shuttle crowding, choose which shuttle is best for their needs, and view detailed shuttle information.



Figure 4.23 User Interface Design: Shuttle Occupancy Details

## Page 2 : Shuttle Check-In Page

This page will allow students to view shuttle crowding estimates at a given time. The app shows the shuttle's routes, hours of operation and frequency, as well as their capacity and an estimate of how many riders are currently utilizing them. The app gives them all of the information about the shuttles on one page.



Figure 4.24 User Interface Design: Digital E-Matric Check-In

### Page 3 : Shuttle Check-In Page

This page allows students to perform a simulated check-in using a digital e-matric card displayed in the app. The check-in action represents boarding the shuttle and is used by the system to simulate an update in shuttle occupancy.

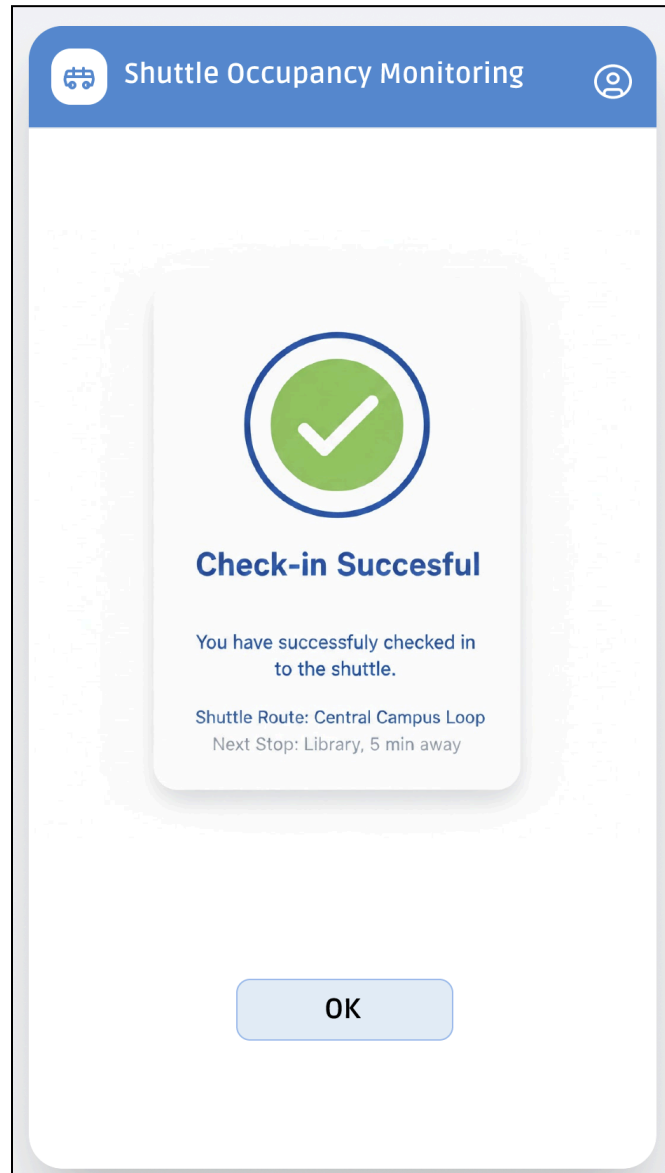
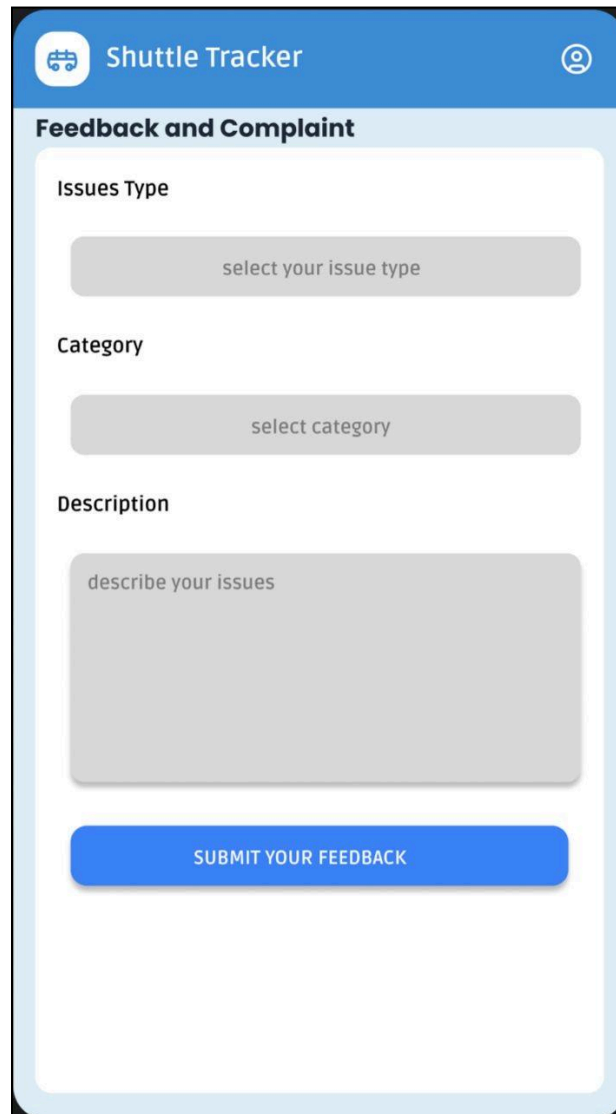


Figure 4.25 User Interface Design: Check-In Confirmation

#### Page 4 : Check-In Confirmation Page

This page displays a confirmation message after a successful check-in, along with basic shuttle information such as route and occupancy status. It provides clear feedback to the student and confirms that the simulated check-in has been recorded by the system.

## User Feedback & Complaint Reporting - Damiya



The image shows a mobile application interface for 'Shuttle Tracker'. At the top is a blue header bar with a white shuttle icon on the left, the text 'Shuttle Tracker' in the center, and a white user profile icon on the right. Below the header is a light blue rounded rectangle containing the title 'Feedback and Complaint' in bold. Under this title are three sections: 'Issues Type' with a grey button labeled 'select your issue type', 'Category' with a grey button labeled 'select category', and 'Description' with a large grey text area labeled 'describe your issues'. At the bottom of this section is a blue button with white text that says 'SUBMIT YOUR FEEDBACK'.

Figure 4.26 User Interface Design: User Feedback & Complaint Reporting

Page 5: The User Feedback and Complaints Reporting sub-module allows users to easily submit feedback, issues, or complaints related to the shuttle service through a simple and intuitive interface. Users can select the issue type and category, provide a detailed description of their concern, and submit it directly within the applicat



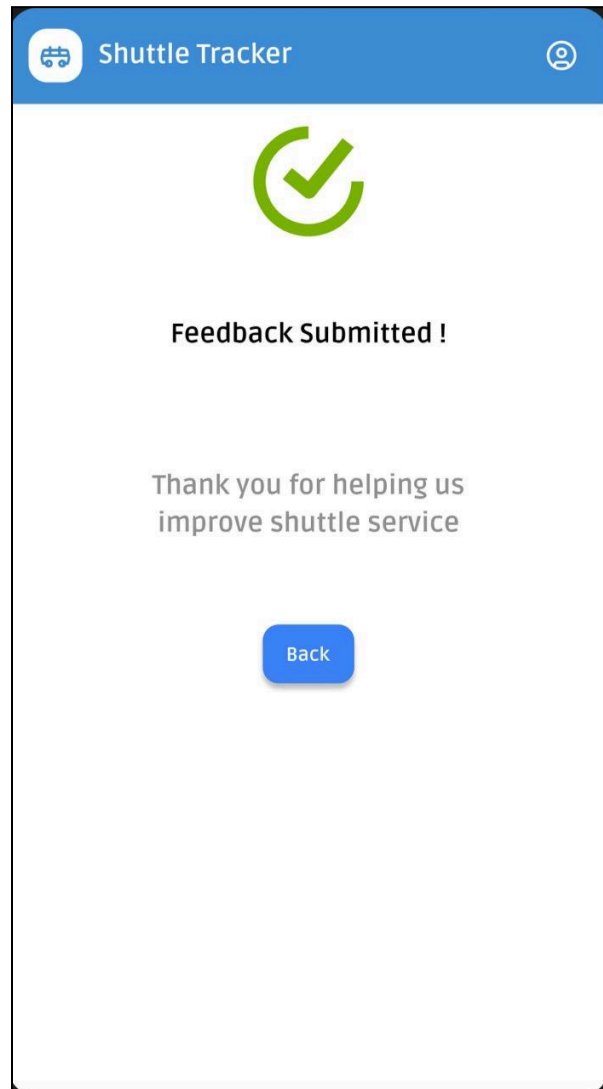


Figure 4.26 User Interface Design: User Feedback Successful Submission

Page 6: Once submitted, the system confirms successful submission and records the feedback for review and action by administrators. This module helps improve service quality by enabling efficient communication between users and service providers, ensuring that problems are tracked, addressed, and resolved in a timely manner while continuously enhancing the overall user experience.

## Transport Analytics Dashboard

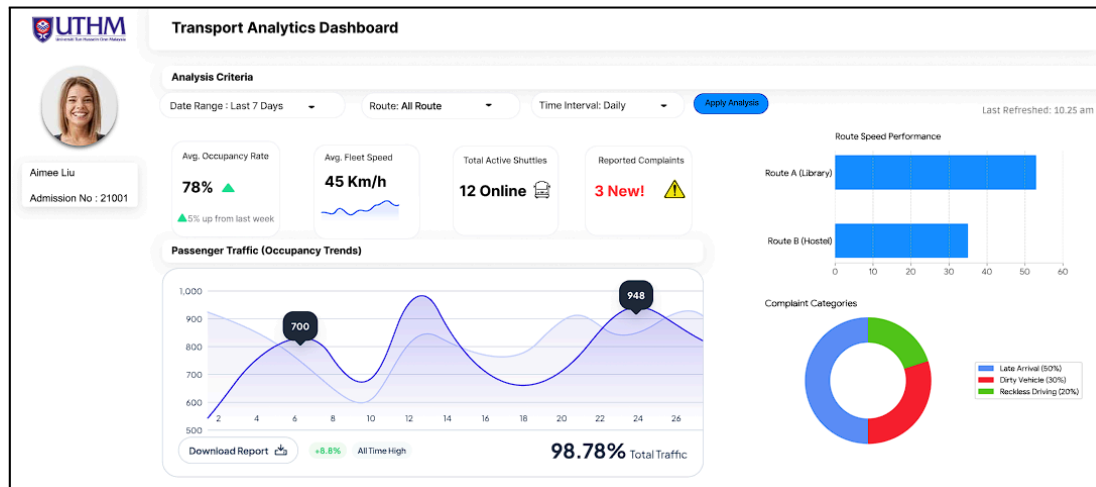


Figure 4.27 User Interface Design: Transport Analytics Dashboard

This dashboard allows admin staff to view and analyze campus shuttle usage and performance at a given time. The system displays transport routes, operational activity, average speed, occupancy levels, and reported complaints in a single interface. It also provides visual summaries such as traffic trends, route speed performance, and complaint categories, enabling administrators to monitor shuttle operations and make informed, data-driven decisions efficiently.

## **4.7 Summary**

The whole Campus Transport and Shuttle Management system for UTHM described the university's organizational structure and identified the main departments involved in managing the system. The current shuttle monitoring system was reviewed, and several weaknesses were identified, such as the lack of passenger count, ETA information, performance analysis, and a proper feedback system. The system requirements have been developed based on these problems in order to enhance the shuttle service's effectiveness, dependability, and usability. The enterprise architecture was introduced to show how business processes, data, and technology are integrated to support real-time tracking, secure access, and better decision-making. Also, as shown in the use case diagram, use case description, sequence diagram, and activity diagram. Besides, ER diagrams are meant to state the entity and relationship that exist in the whole architecture.

## **CHAPTER 5 : CONCLUSION**

### **5.1 Introduction**

This chapter concludes the project by summarizing the overall outcomes, contributions, and limitations of the proposed Campus Transport and Shuttle Management System for Universiti Tun Hussein Onn Malaysia (UTHM). It highlights how the system addresses existing operational challenges through the application of Enterprise Architecture principles and digital technologies. In addition, this chapter discusses the system's achievements, identified constraints, and potential future enhancements, providing a comprehensive reflection on the effectiveness of the proposed solution and its role in supporting UTHM's smart campus initiative.

### **5.2 System Contribution / Achievements**

#### **What the system achieved**

The Campus Transport & Shuttle Management System successfully integrates six key modules to enhance both student convenience and administration. It provides clear information on shuttle routes, operating hours and stops through the Route & Schedule Display module, while the Live Shuttle Tracking and ETA Predictor modules allow students to monitor buses in real time and view estimated arrival times. The Shuttle Occupancy Monitoring module simulates bus capacity levels to help students track the occupancy, and the User Feedback & Complaint Reporting module enables students to submit issues, which administrators can review and resolve efficiently. Additionally, the Transport Analytics Dashboard offers visual insights on peak hours, route usage, and complaints. By integrating student profiles from the Student Management Portal (SMP), the system centralizes shuttle information and enhances overall user experience.

## **How it helps the organization**

The suggested integrated enterprise system improves UTHM's organizational efficiency by moving from fragmented, manual processes like Google Forms and Excel sheets to a centralized digital solution. This transfer efficiently reduces operational silos and data redundancy by linking the transport module to the university's master database (SMP/SMS), assuring data integrity and consistency across departments. The system provides administrators with a Transport Analytics Dashboard, which converts real-time data into actionable insights about shuttle usage trends and peak hours, allowing for more informed and efficient decision-making. Finally, this architecture serves as a scalable template for UTHM's "Smart Campus" goal, enhancing service quality and infrastructure management while ensuring long-term interoperability with future digital transformation initiatives.

### 5.3 System Constraints

Despite the benefits and contributions of the proposed Campus Transport & Shuttle Management System, several constraints were identified during the design and development of the system.

One of the main constraints is the use of simulated data for shuttle location, occupancy, and ETA calculations. Since the system is developed as a prototype using Figma, real-time integration with physical GPS devices, passenger counting sensors, or actual shuttle hardware was not implemented. As a result, the accuracy of live tracking, occupancy levels, and arrival time predictions is limited and may not fully represent real-world shuttle operations.

The system is also constrained by **limited system integration** with existing UTHM transport infrastructure. Although the design supports integration with legacy systems such as the Student Management Portal (SMP) and Staff Management System (SMS) for authentication purposes, the actual integration with existing shuttle tracking systems like Katsana was not implemented in this prototype. This limits the system's ability to function as a fully operational enterprise system.

In addition, the system relies heavily on **stable internet connectivity**. Since the application is cloud-based and accessed through mobile devices and web browsers, any network disruptions may affect system availability, real-time updates, and user experience. This may be a challenge in areas of the campus with weak network coverage.

Overall, these constraints highlight areas for improvement and future enhancement, while still allowing the system to effectively demonstrate how Enterprise Architecture and digital solutions can improve campus transport management at UTHM.

## **5.4 Future Suggestions**

For future improvements, the Campus Transport & Shuttle Management System can be enhanced through the use of AI-based demand prediction. It is to optimize shuttle schedules based on usage trends, academic calendars, and peak hours. Developing a dedicated mobile application with push notifications would further improve student convenience. It helps by providing real-time alerts on delays, route changes, and shuttle occupancy. These enhancements would strengthen operational efficiency, support UTHM's Smart Campus initiative, and ensure the system remains scalable and adaptable to future campus transportation needs.

## **5.5 Summary**

This chapter concludes the project by assessing the overall effectiveness of the proposed Campus Transport and Shuttle Management System for Universiti Tun Hussein Onn Malaysia (UTHM). The system demonstrates how Enterprise Architecture principles and digital technologies can be applied to address existing campus transportation challenges. By integrating modules such as route and schedule display, live shuttle tracking, ETA prediction, occupancy monitoring, user feedback, and a transport analytics dashboard, the system improves student convenience and centralizes shuttle information while supporting data-driven decision-making for administrators.

Despite these achievements, several limitations were identified, including the use of simulated data, limited integration with existing shuttle infrastructure, and dependency on stable internet connectivity. Nevertheless, the system effectively showcases the potential of a centralized enterprise solution in supporting UTHM's Smart Campus initiative. Future enhancements such as AI-based demand prediction and a dedicated mobile application with real-time notifications can further improve operational efficiency, scalability, and overall user experience.

Video Presentation - <https://youtu.be/GTU8LRXz9Lw>

Figma Prototype -

<https://www.figma.com/proto/i0FbD4WDwbJoNOPb6bVvUz/GROUP-5---Campus-Transport---Shuttle-Management?node-id=1-8483&t=vJT5DB1KkxNqJaWS-1>



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